

Measurement of Absolute Double Differential Cross-Section in Invariant Mass and p_T Bins

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April 29, 2026

1 Input Files and Bin Ranges

Input Data Files:

- Merged ROOT files containing reconstruction and hodoscope efficiencies.
- **Roadsets:** RS57, RS59, RS62, RS67 (run 3089), and RS70.
- **Targets:** Liquid Hydrogen (LH2), Liquid Deuterium (LD2), and Empty Flask.

Kinematic Binning Definition:

- **Mass Bins (GeV):** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.7]
- **p_T Bins (GeV):** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]
- **x_F Bins:** [0.0, 0.85] with a fixed step width of 0.05.

2 Event Selection Criteria

Dimuon Kinematics:

- $4.2 < Mass < 8.8$ GeV, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos \theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

Track Quality & Acceptance:

- $\chi_{target}^2 < 15$, $\chi_{dimuon}^2 < 18$, and $\chi^2/(nHits - 5) < 12$.
- Hit requirements: $nHits > 13$ per track, $nHits_1 + nHits_2 > 29$, St1 total > 8 .
- Track vertex longitudinal limits: $-320 < z_v < -5$.

- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for $\text{runID} \geq 11000$ and 0.4 otherwise.

Detector Occupancy Limits:

- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.

3 Plots and Distributions

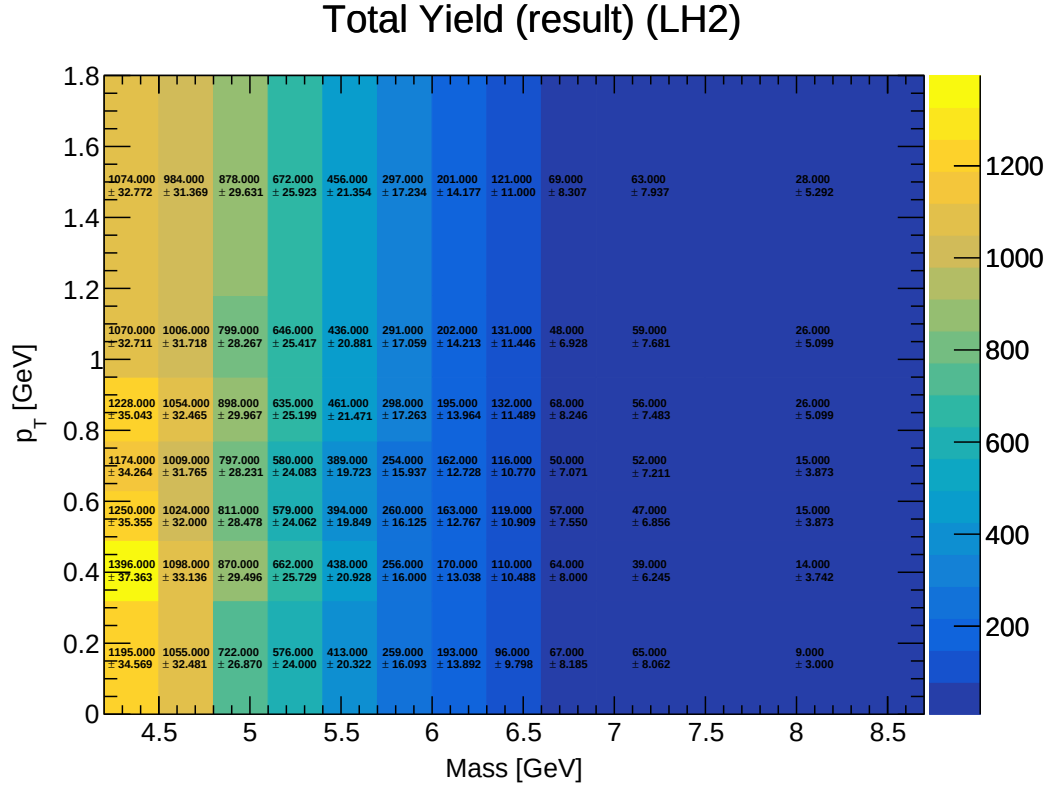


Figure 1: Y_total_LH2

Mix Yield (result_mix) (LH2)

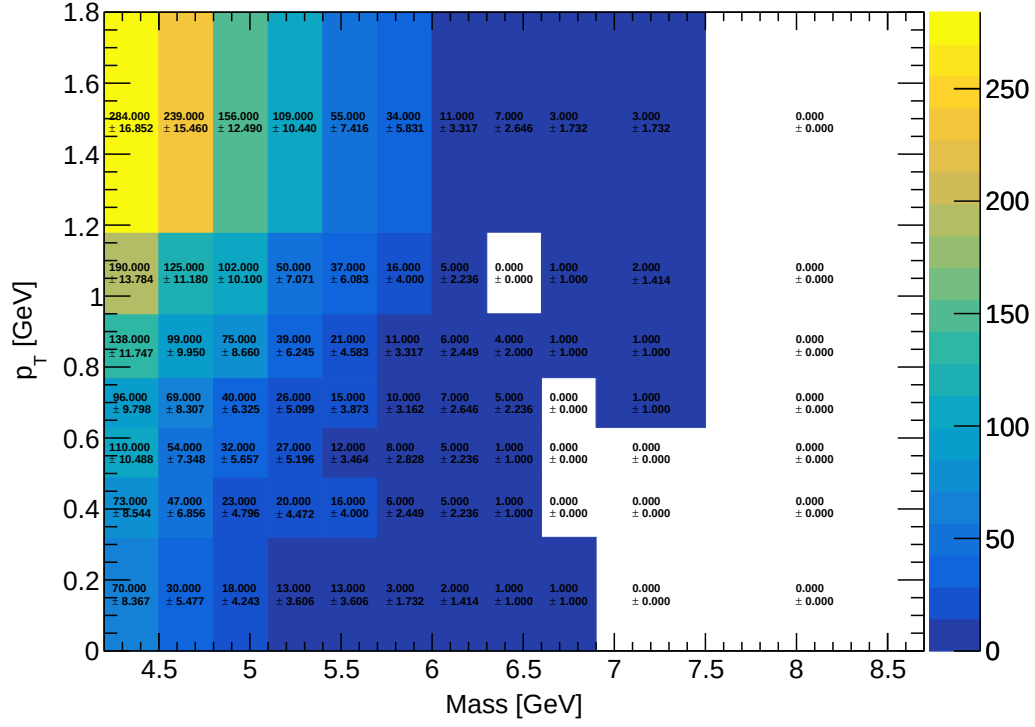


Figure 2: Y_mix_LH2

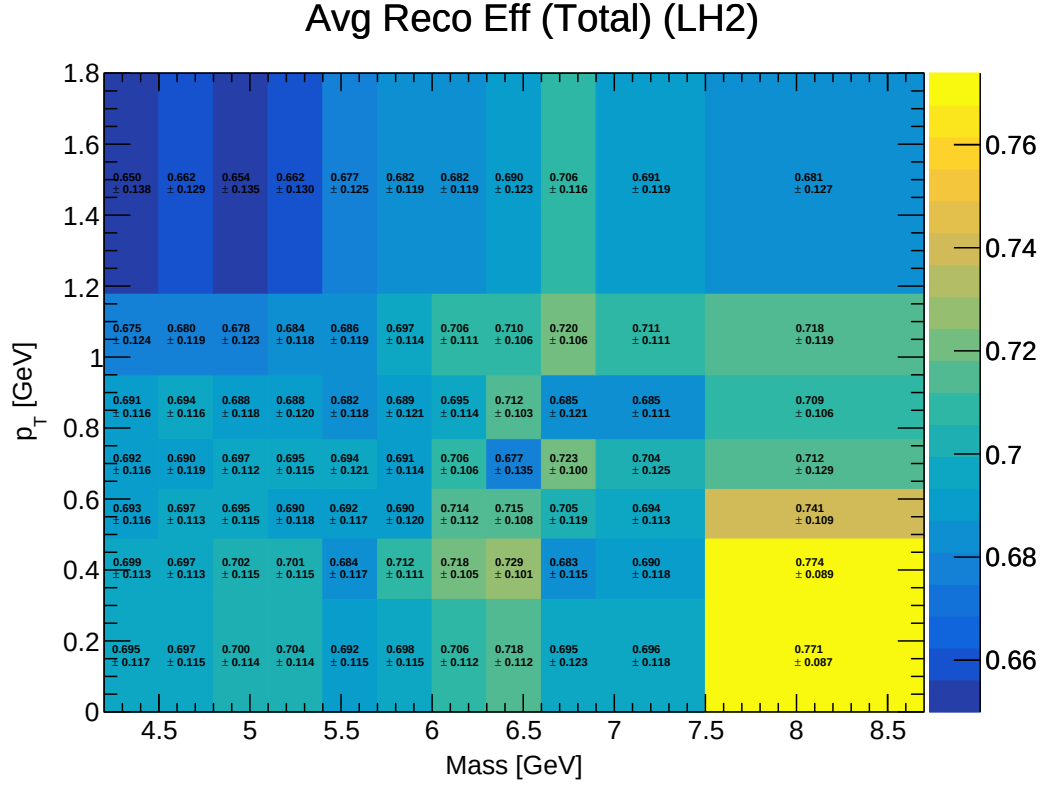


Figure 3: $E_{\text{total_reco_LH2}}$

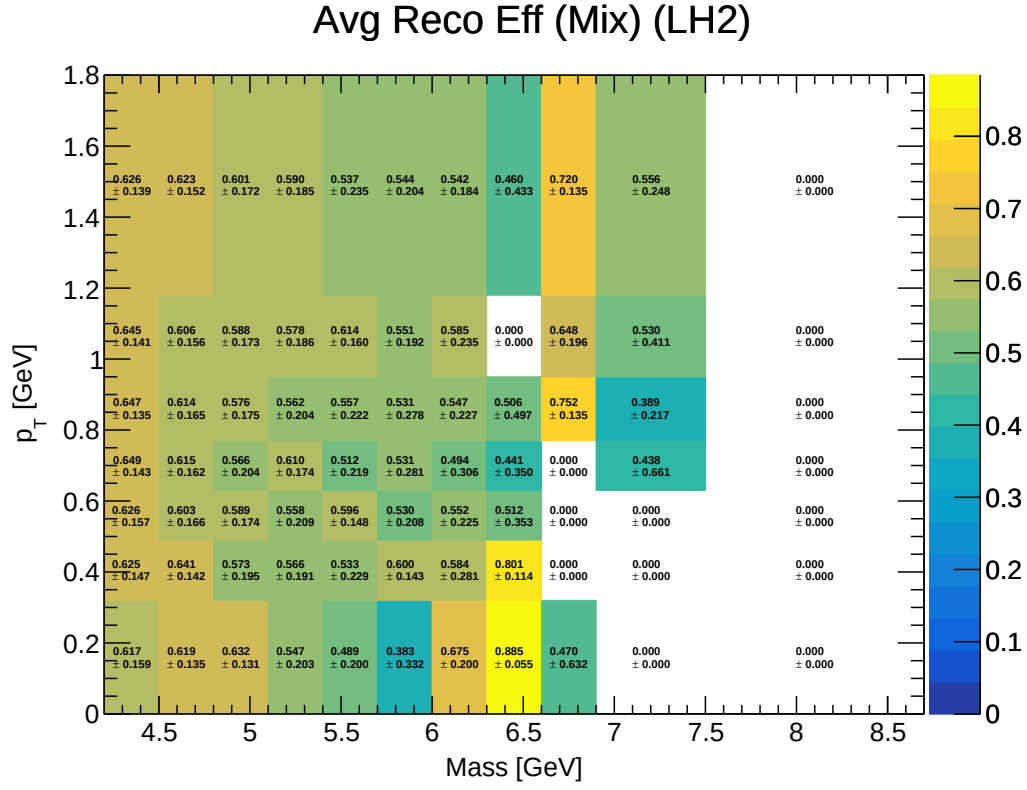


Figure 4: E.mix_reco.LH2

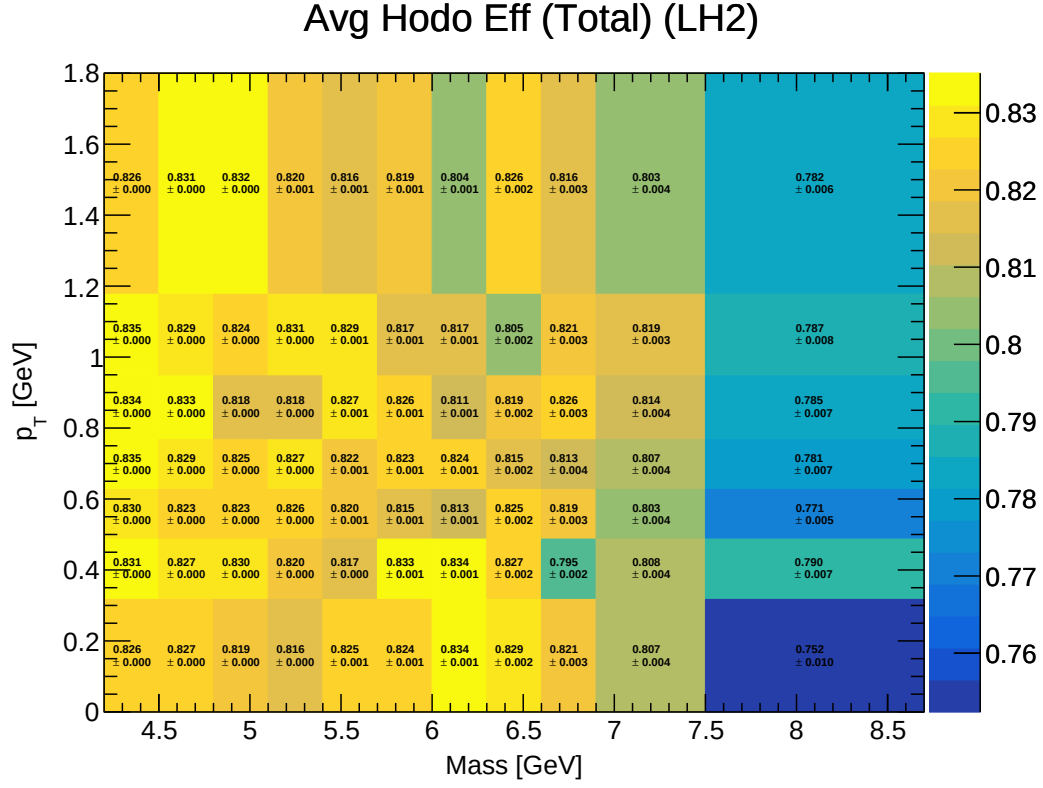


Figure 5: E_total_hodo_LH2

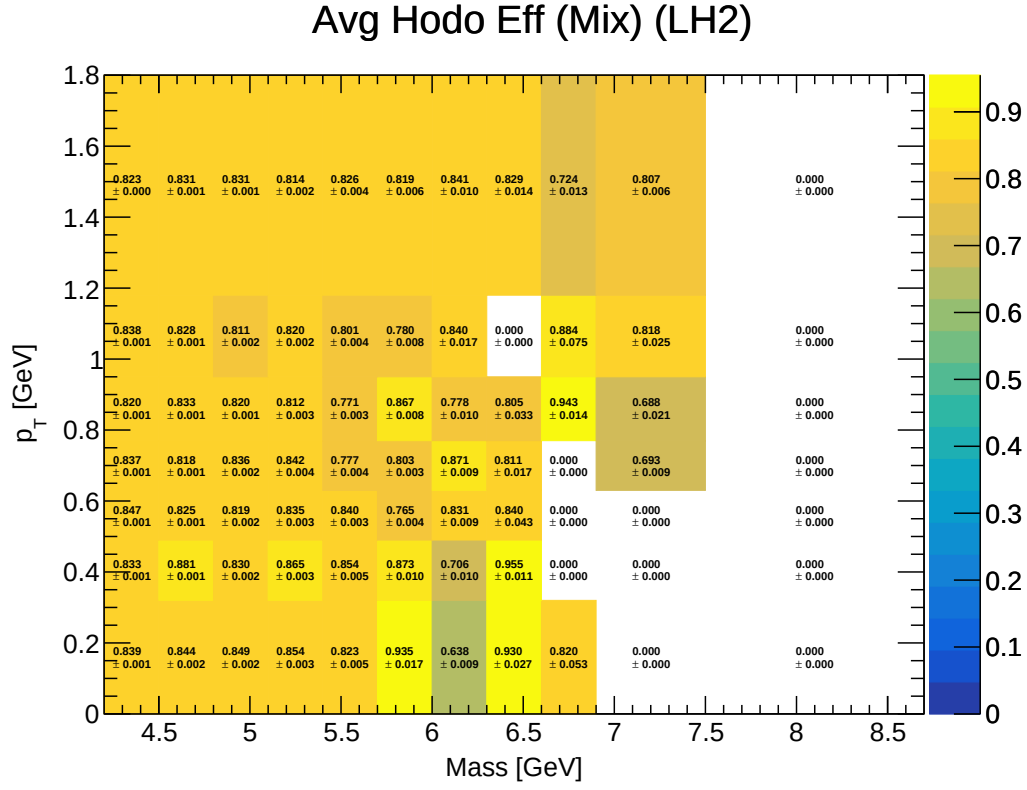


Figure 6: E_mix_hodo_LH2

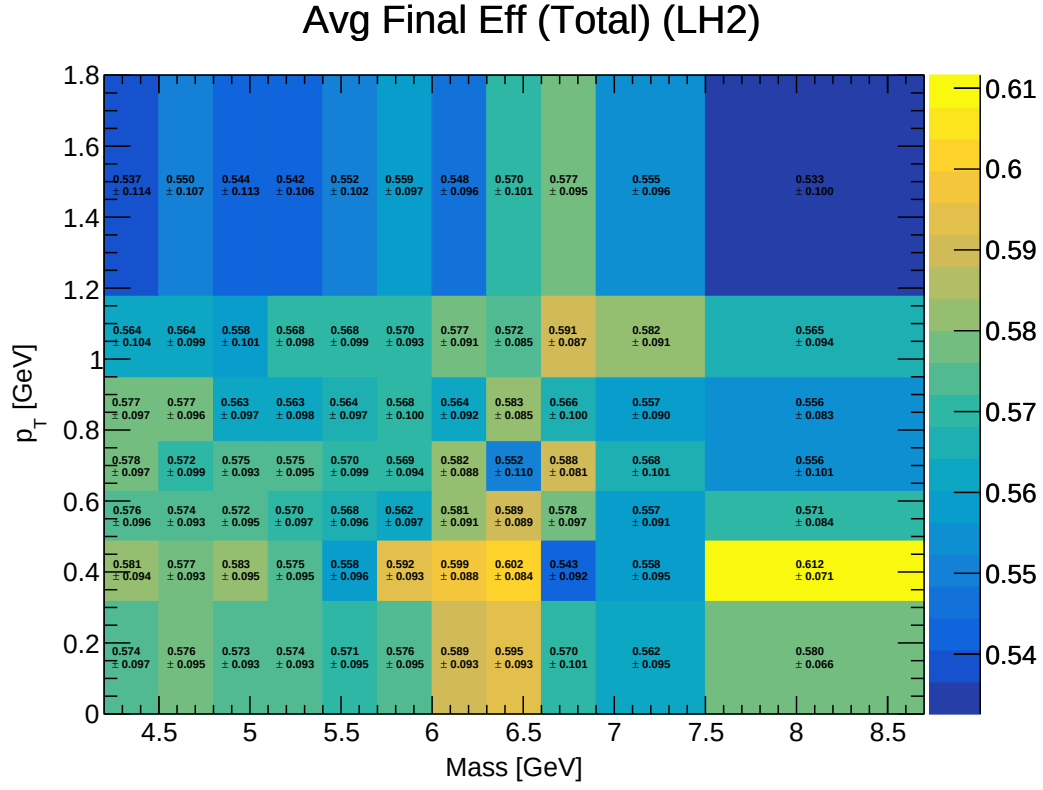


Figure 7: E_total_final_LH2

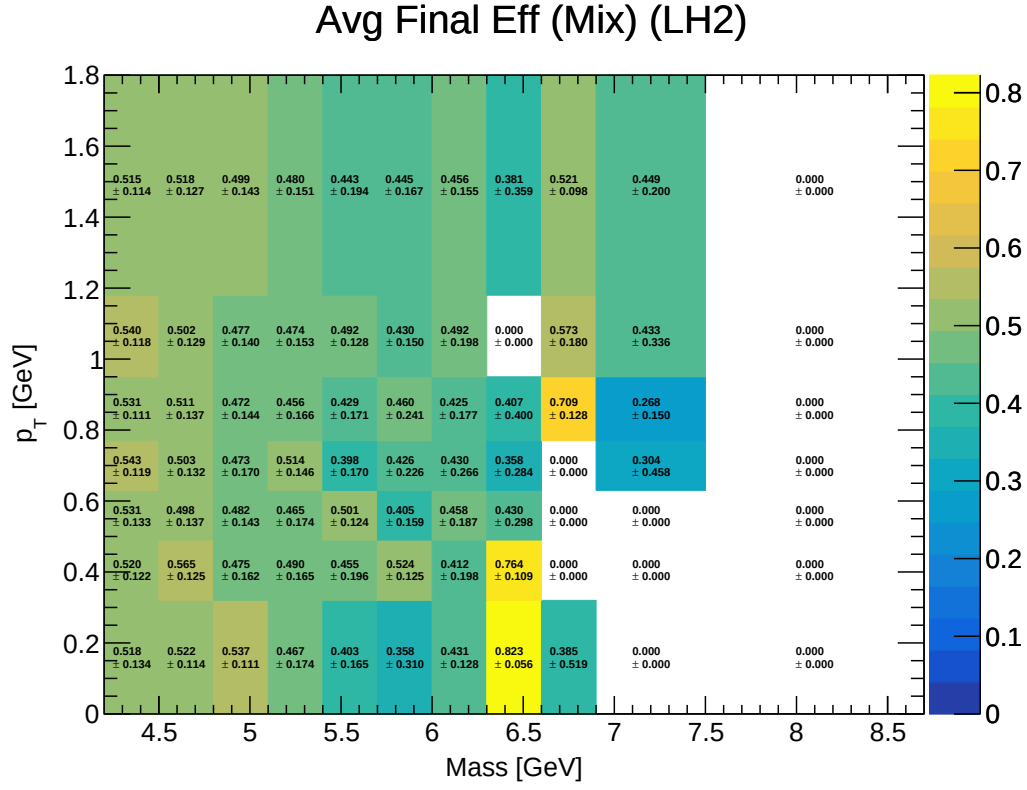


Figure 8: E_{mix}_final_LH2

Avg Signal Efficiency (LH2)

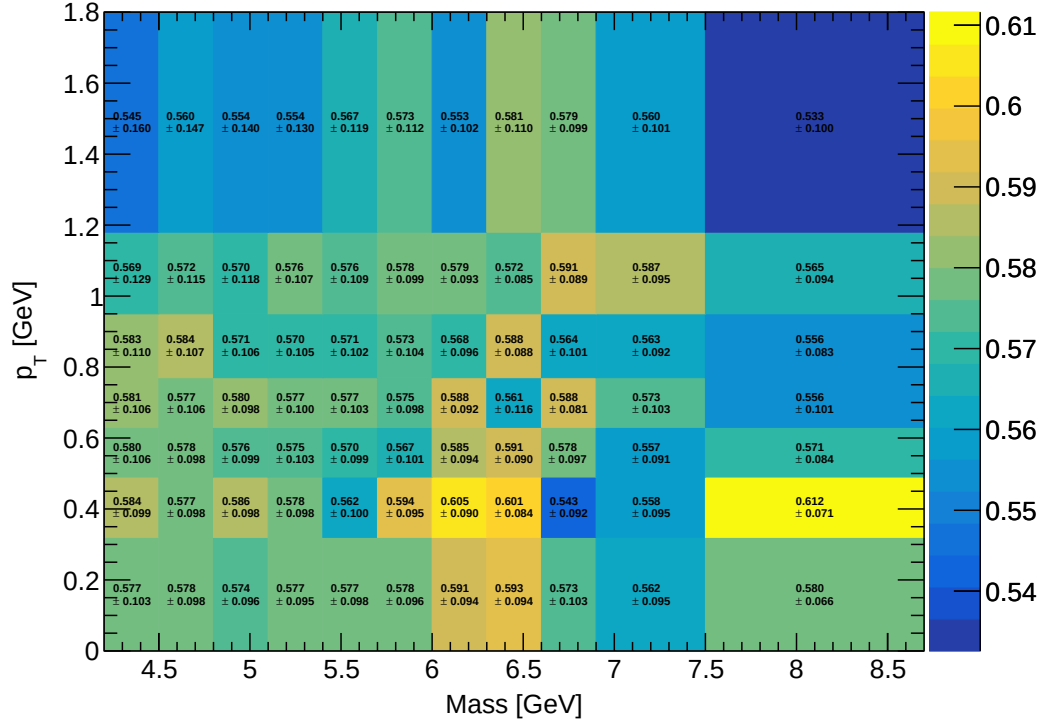


Figure 9: E_final_signal_LH2

Corrected Yield (Total Error) (LH2)

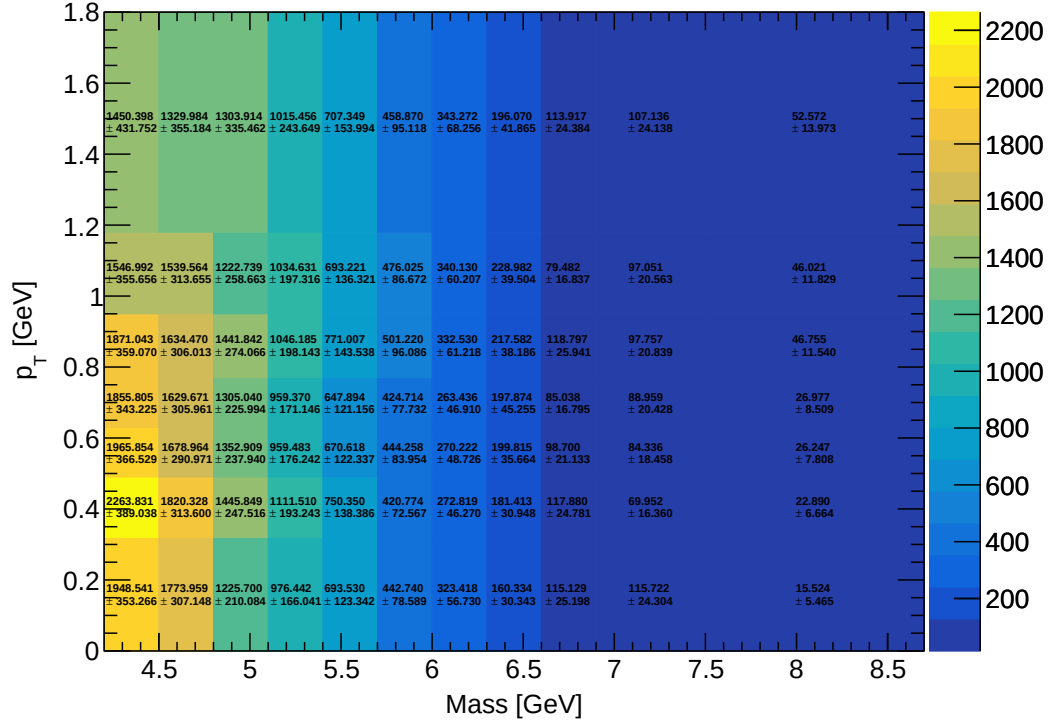


Figure 10: $Y_{\text{corrected_LH2}}$

Total Yield (result) (LD2)

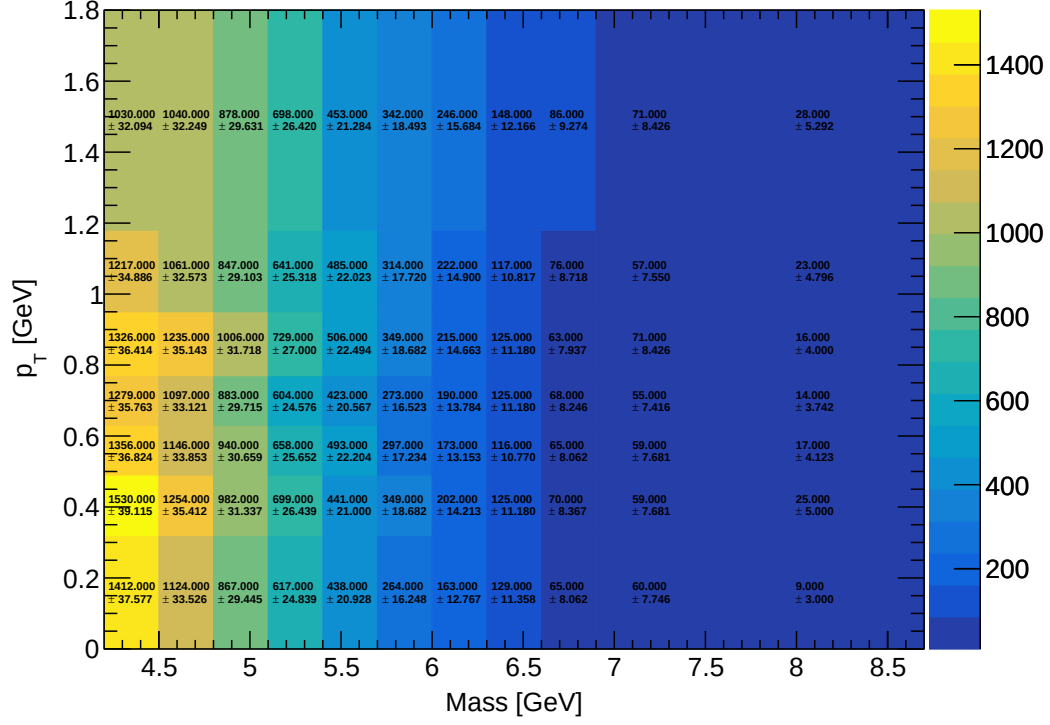


Figure 11: Y_{total_LD2}

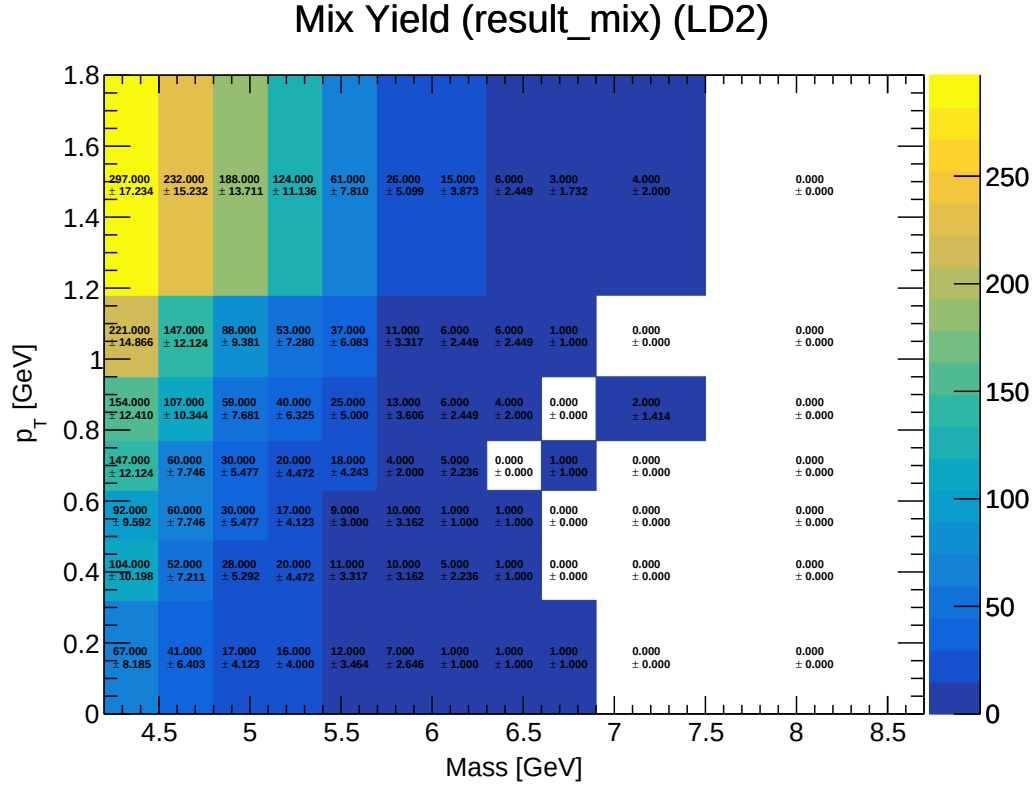


Figure 12: Y_mix_LD2

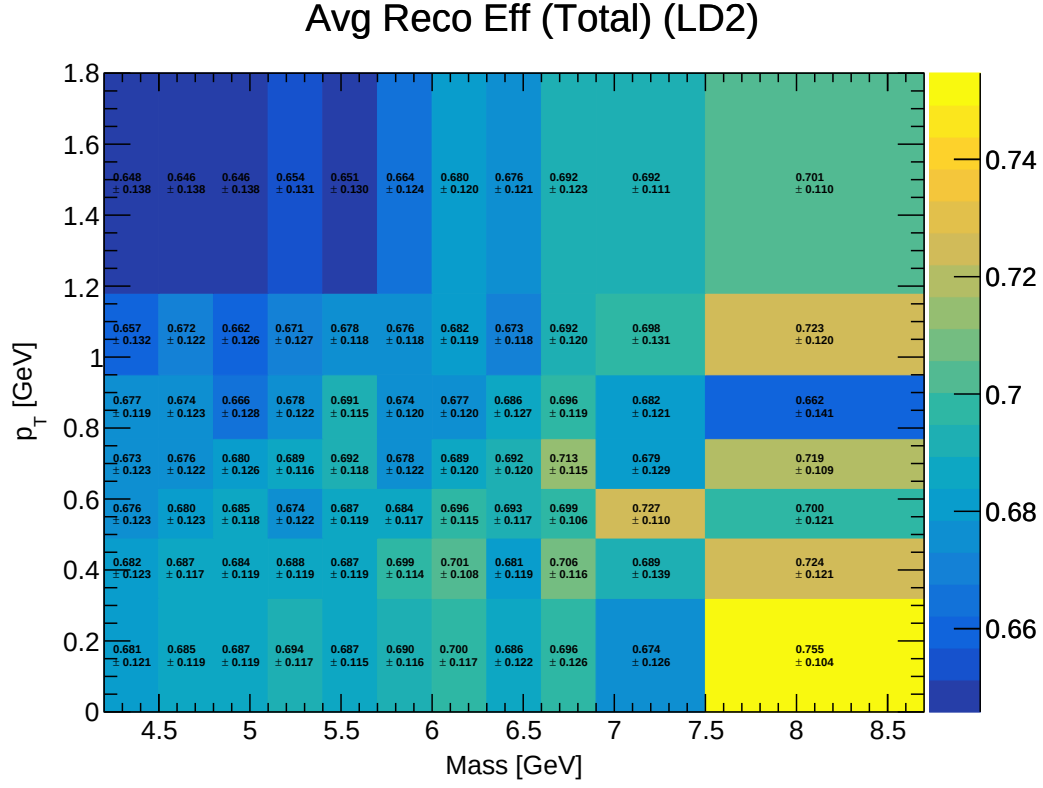


Figure 13: E.total_reco_LD2

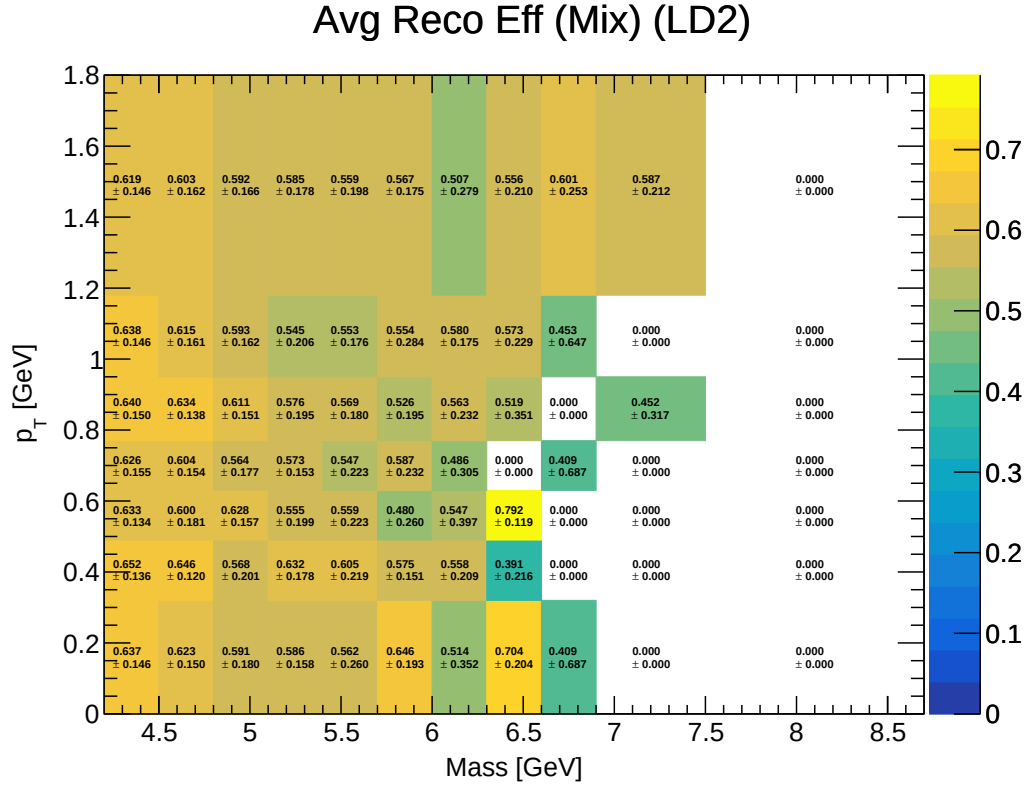


Figure 14: E_mix_reco_LD2

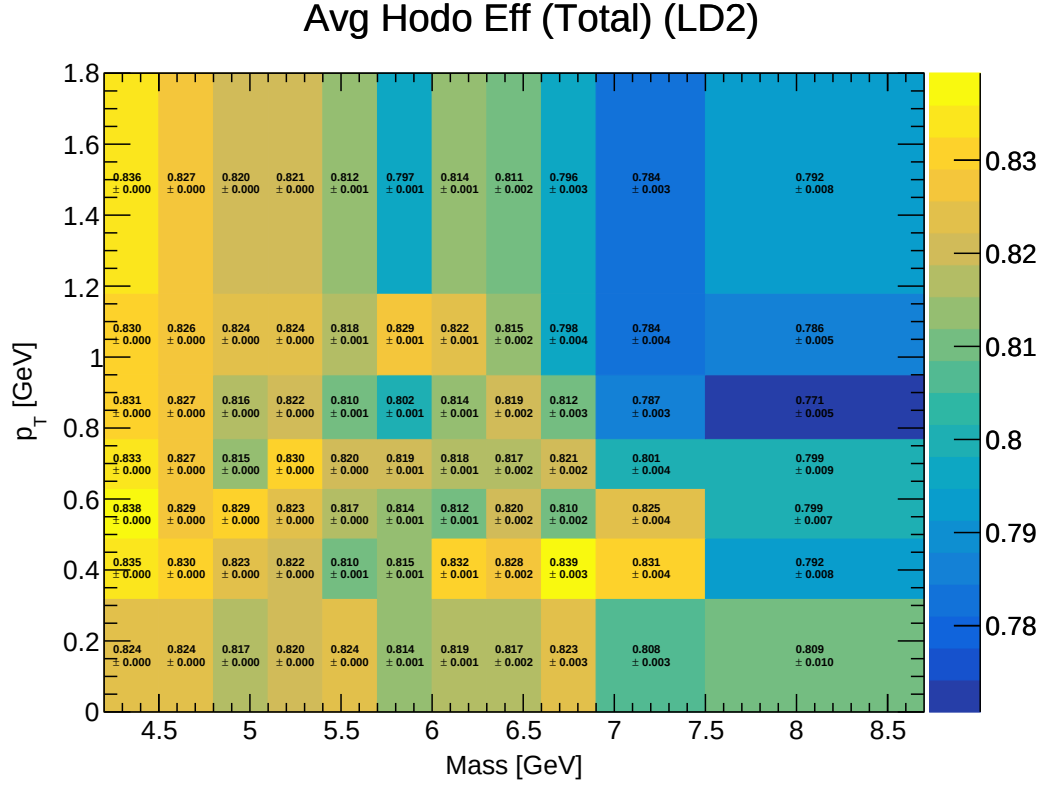


Figure 15: E_{total_hodo_LD2}

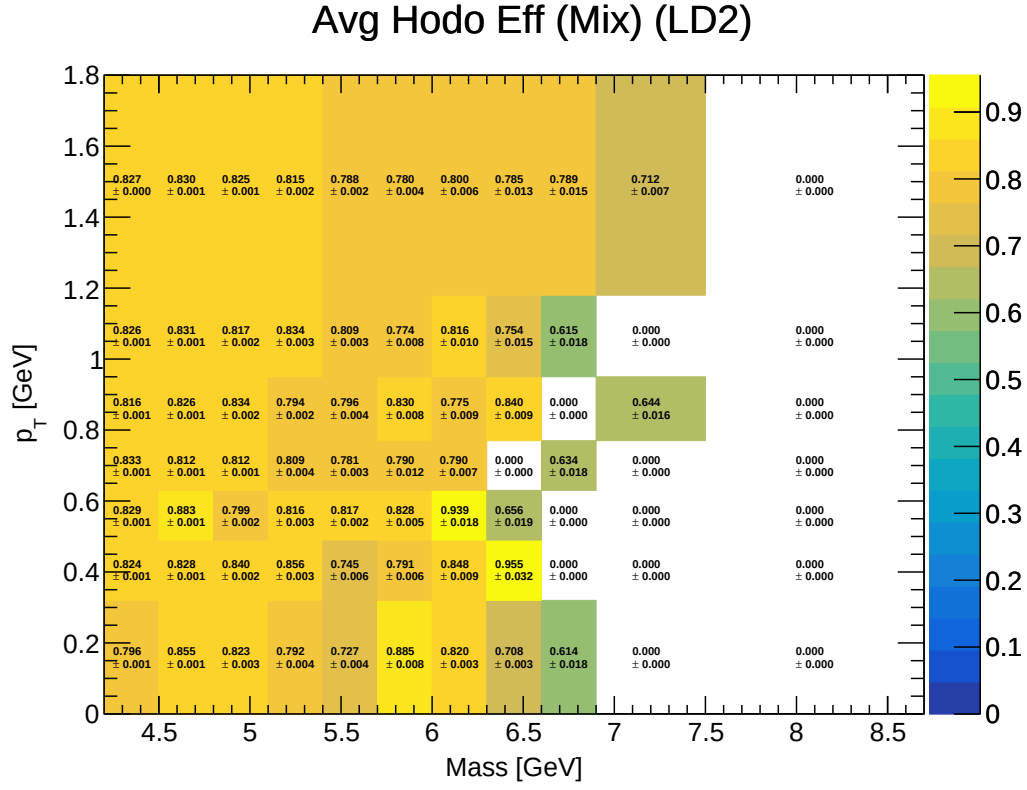


Figure 16: E.mix.hodo_LD2

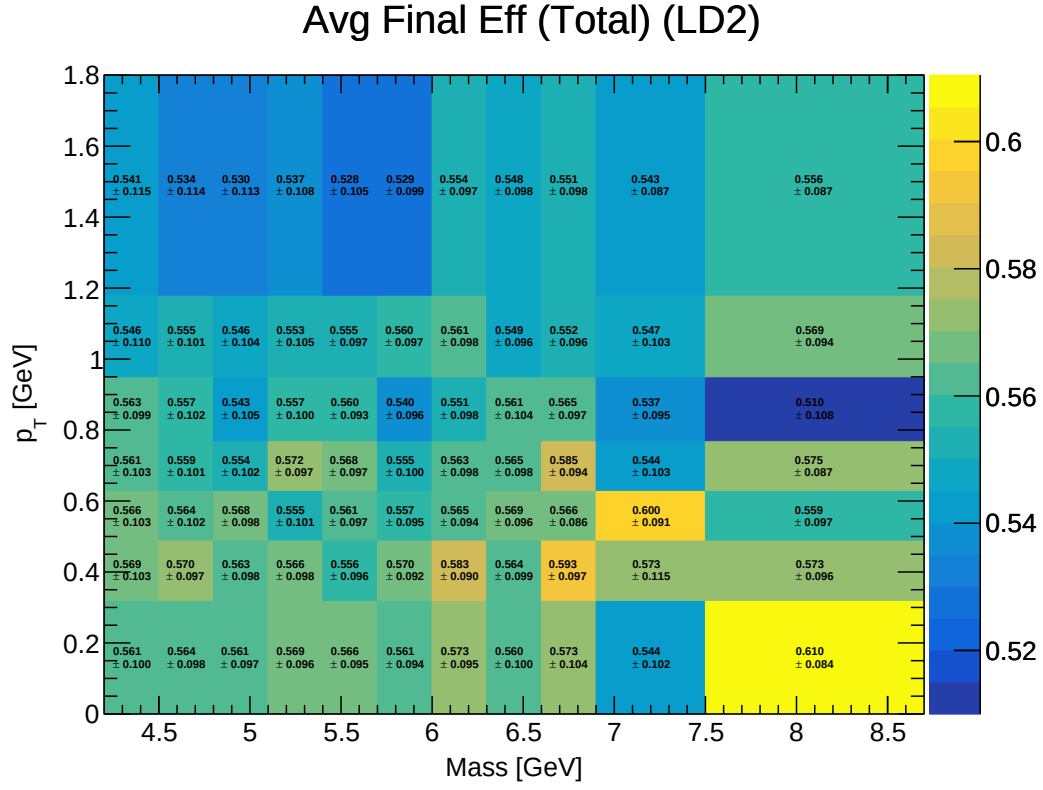


Figure 17: E_total_final_LD2

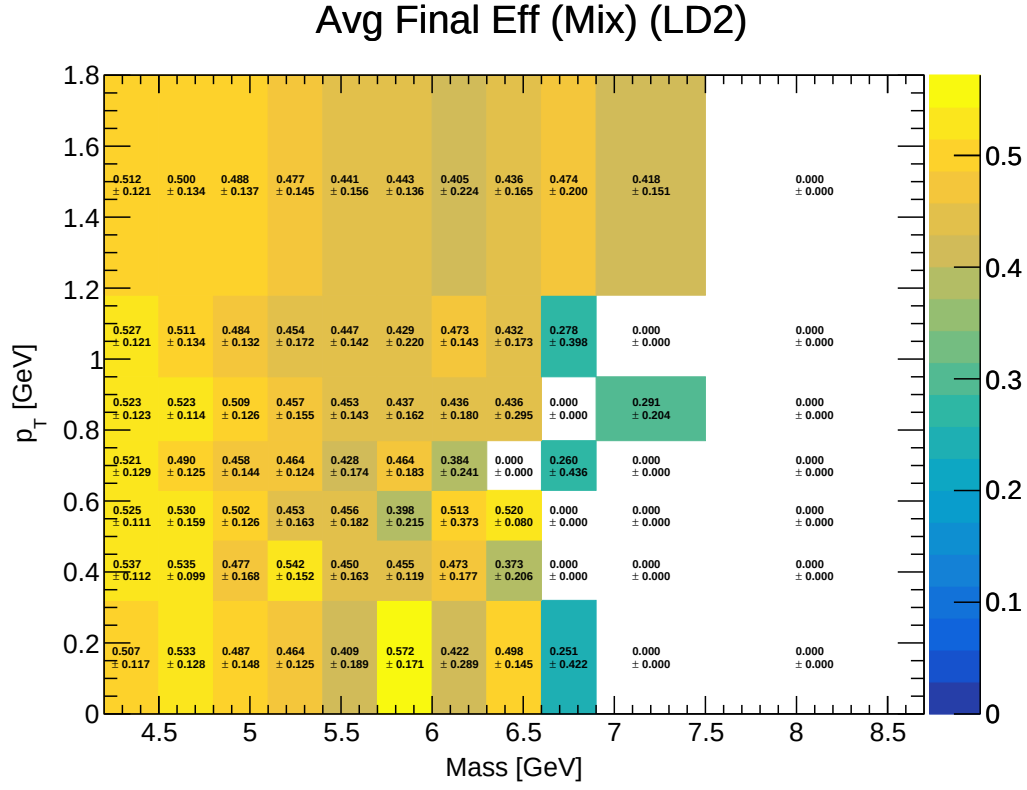


Figure 18: E_mix_final_LD2

Avg Signal Efficiency (LD2)

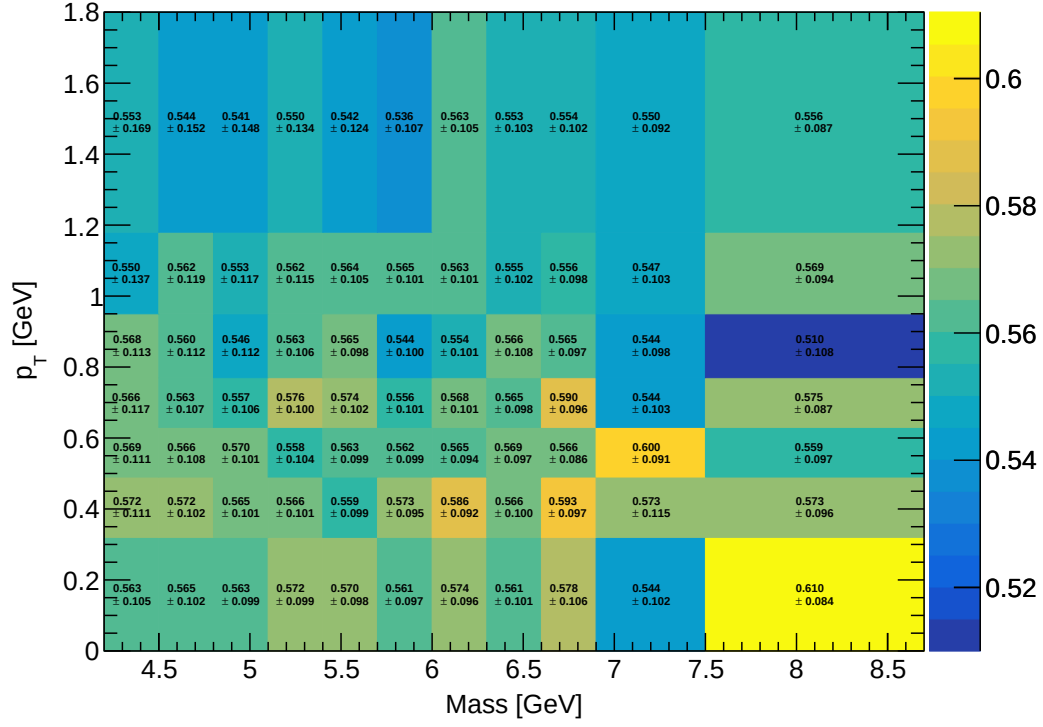


Figure 19: E_final_signal_LD2

Corrected Yield (Total Error) (LD2)

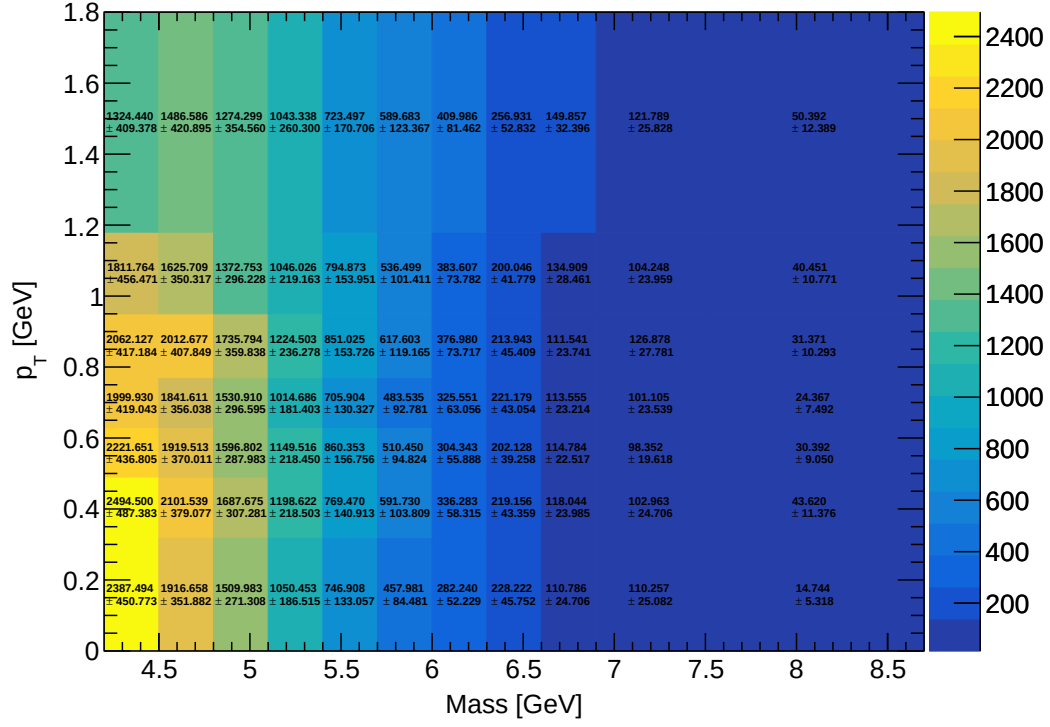


Figure 20: $Y_{\text{corrected_LD2}}$

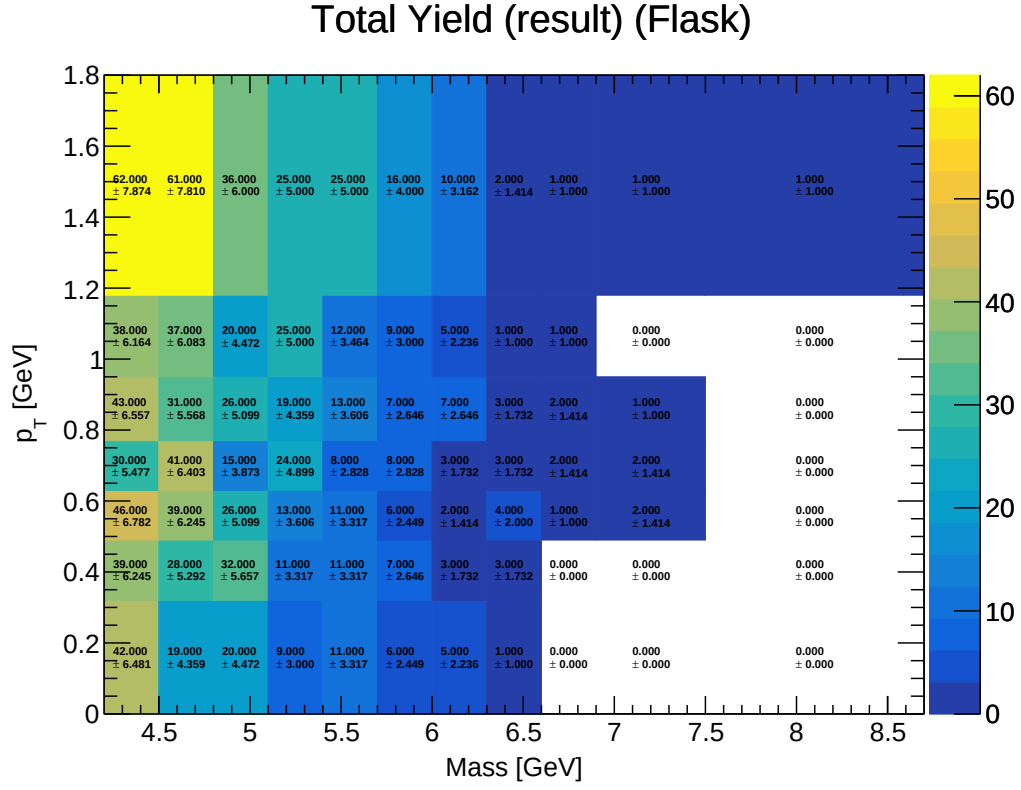


Figure 21: $Y_{\text{total_Flask}}$

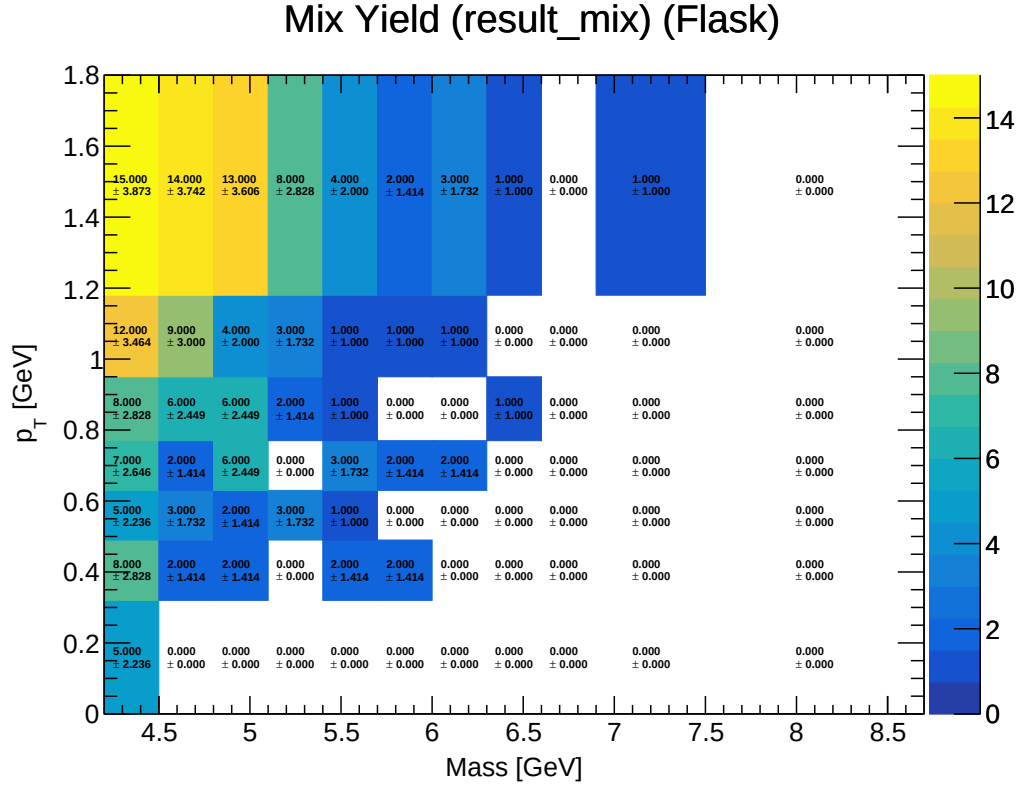


Figure 22: Y_mix_Flask

Avg Reco Eff (Total) (Flask)

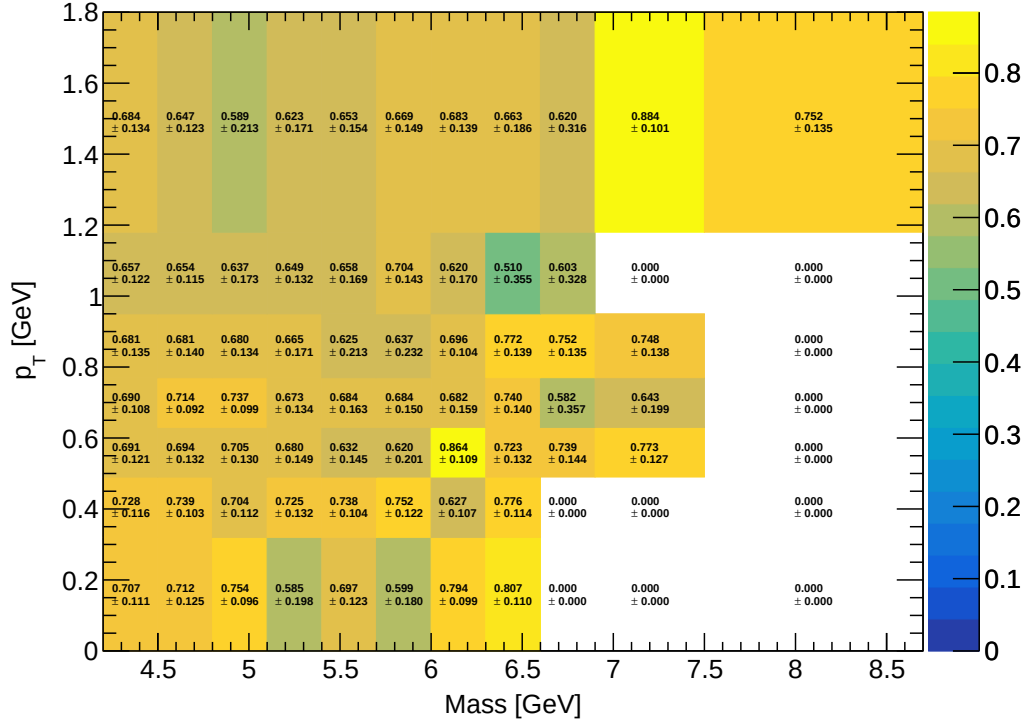


Figure 23: E_total_reco_Flask

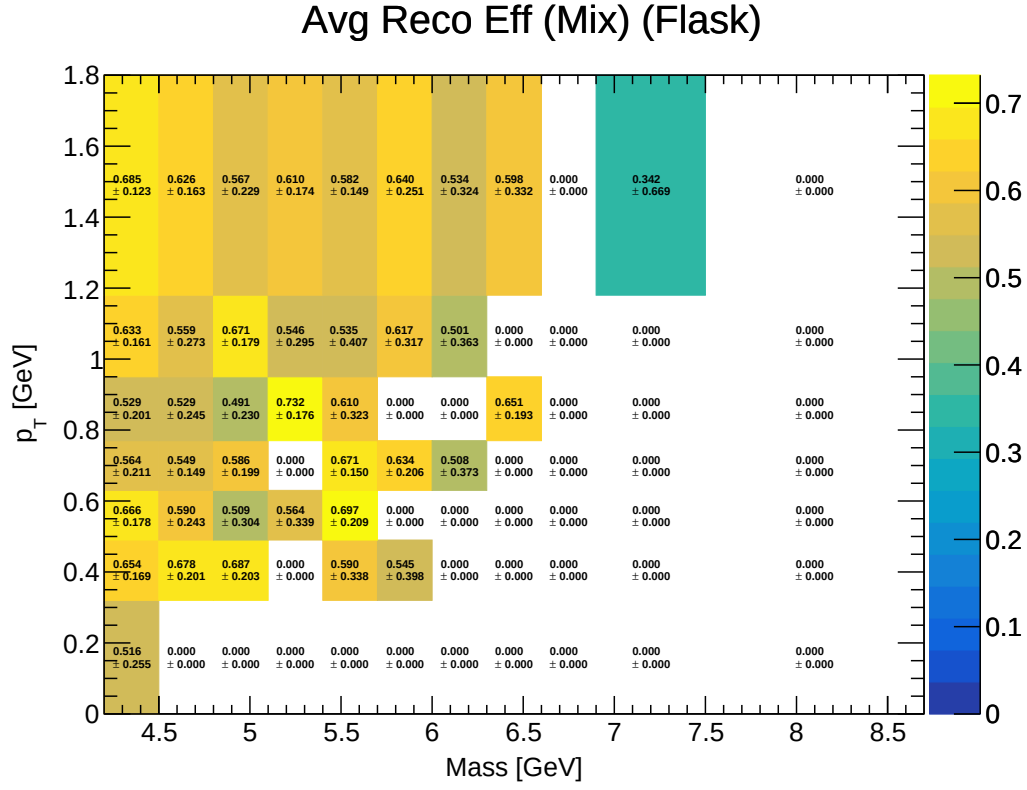


Figure 24: E.mix_reco.Flask

Avg Hodo Eff (Total) (Flask)

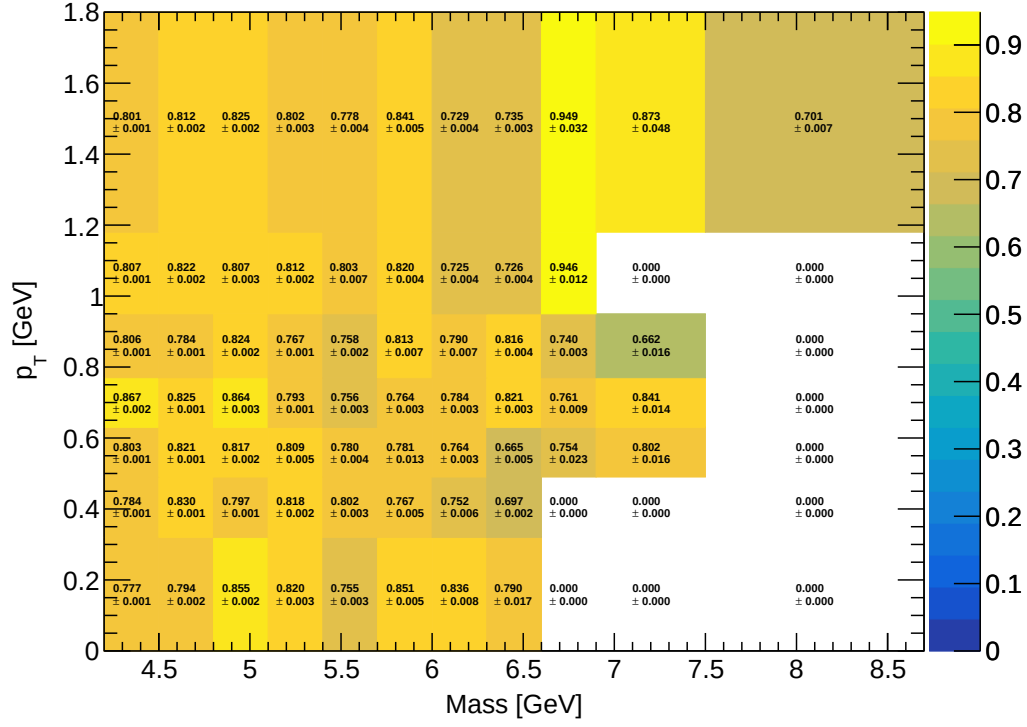


Figure 25: E_total_hodo_Flask

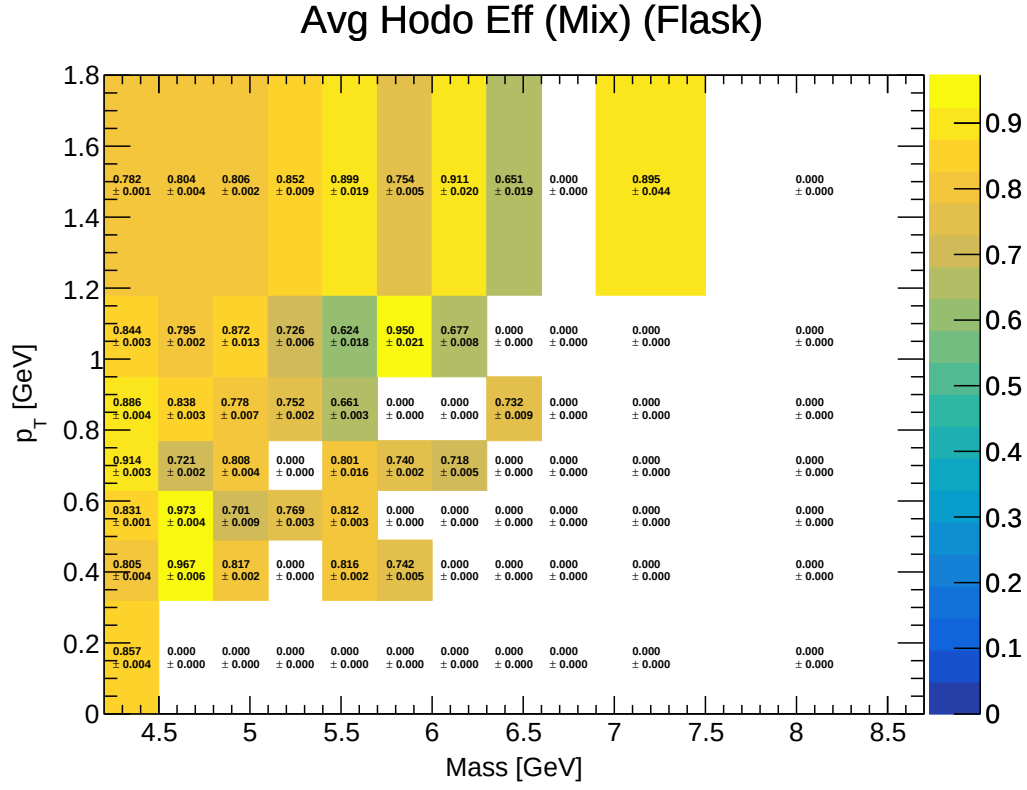


Figure 26: E_mix_hodo_Flask

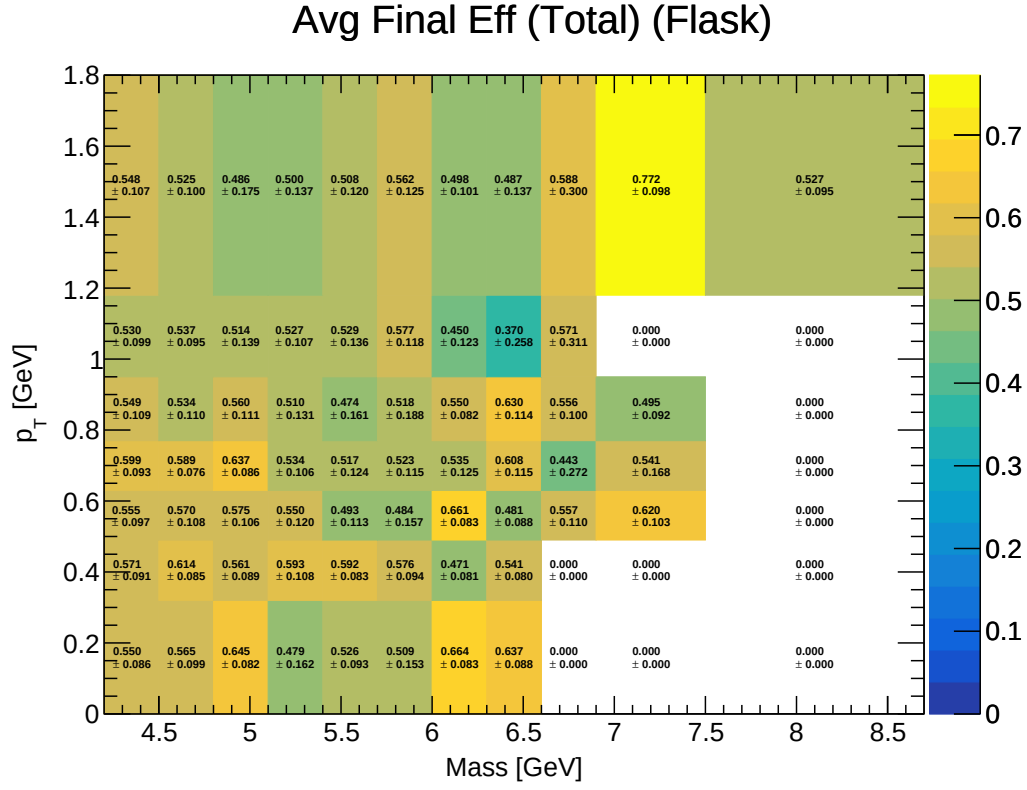


Figure 27: E_total_final_Flask

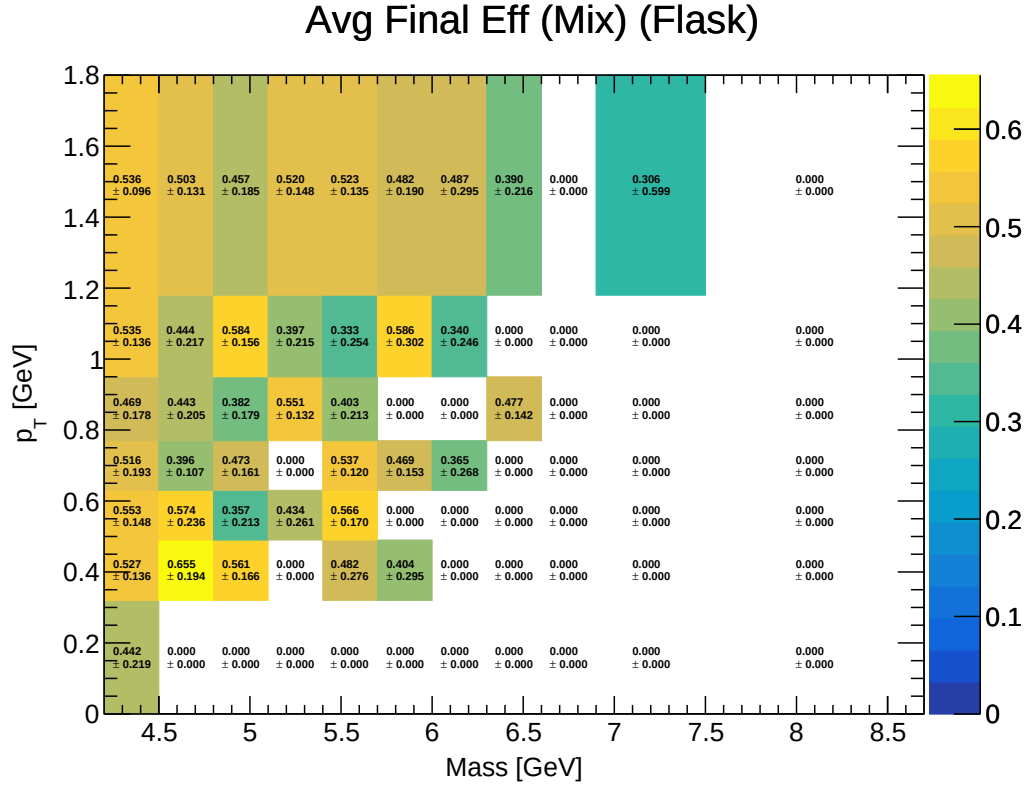


Figure 28: $E_{mix_final_Flask}$

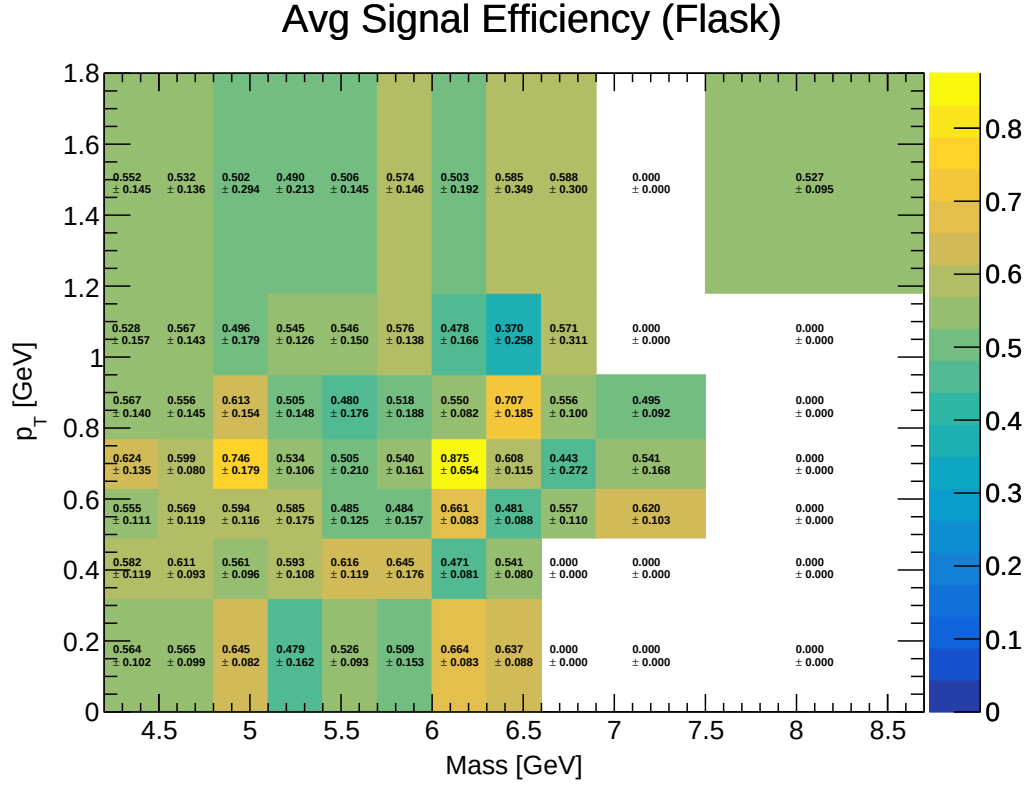


Figure 29: E_final_signal_Flask

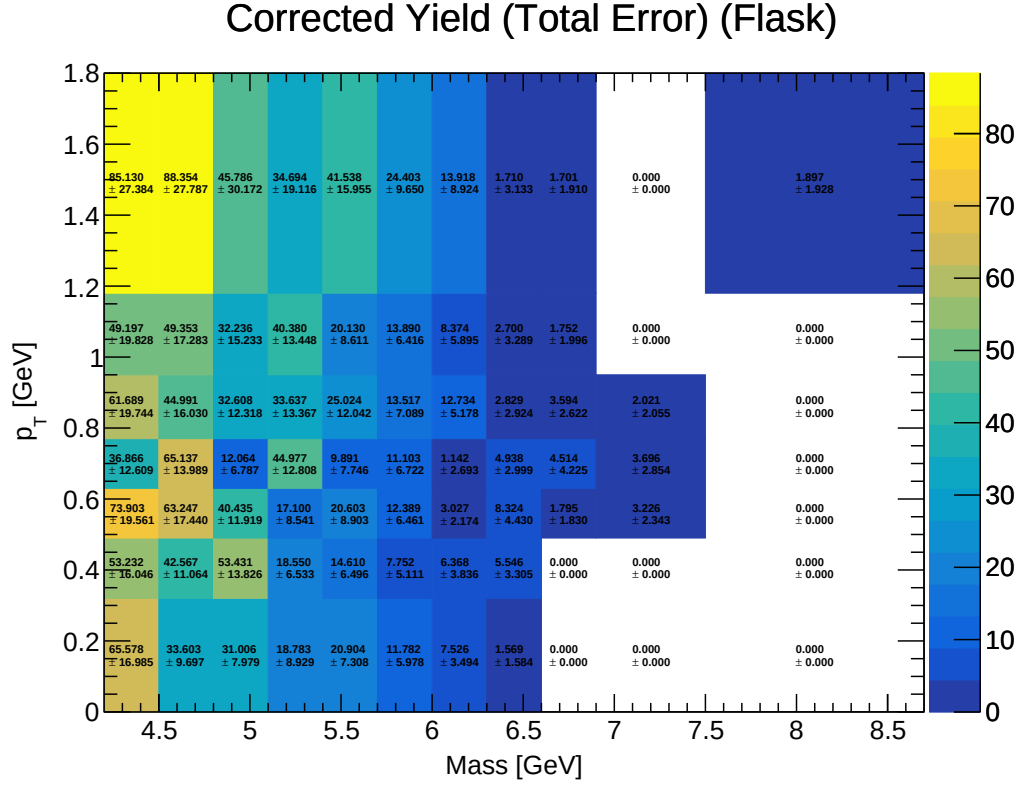


Figure 30: $Y_{\text{corrected_Flask}}$

Corrected Yield (LH2 - Flask)

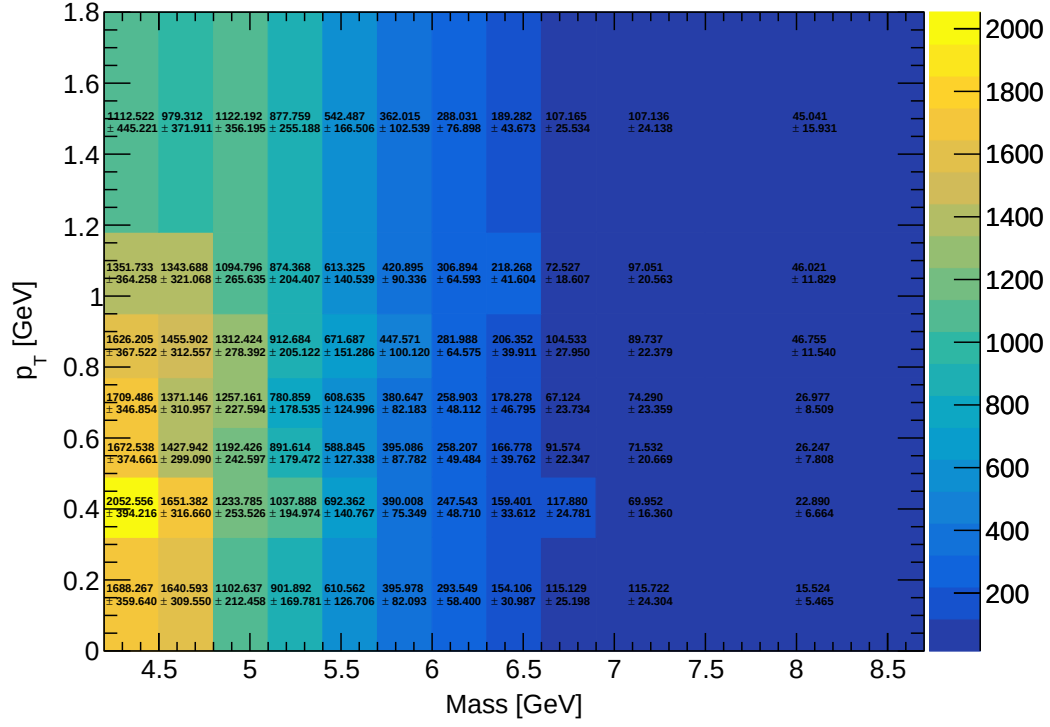


Figure 31: Y_corrected_Subtracted_LH2

Corrected Yield (LD2 - LH2 - Flask)

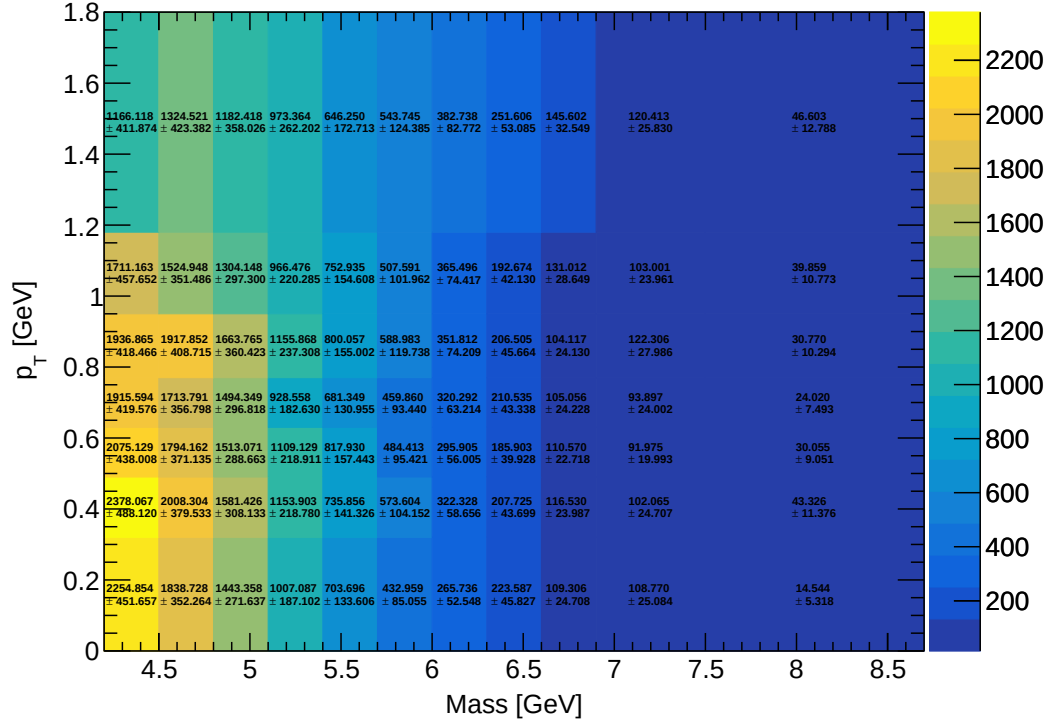


Figure 32: Y_corrected_Subtracted_LD2

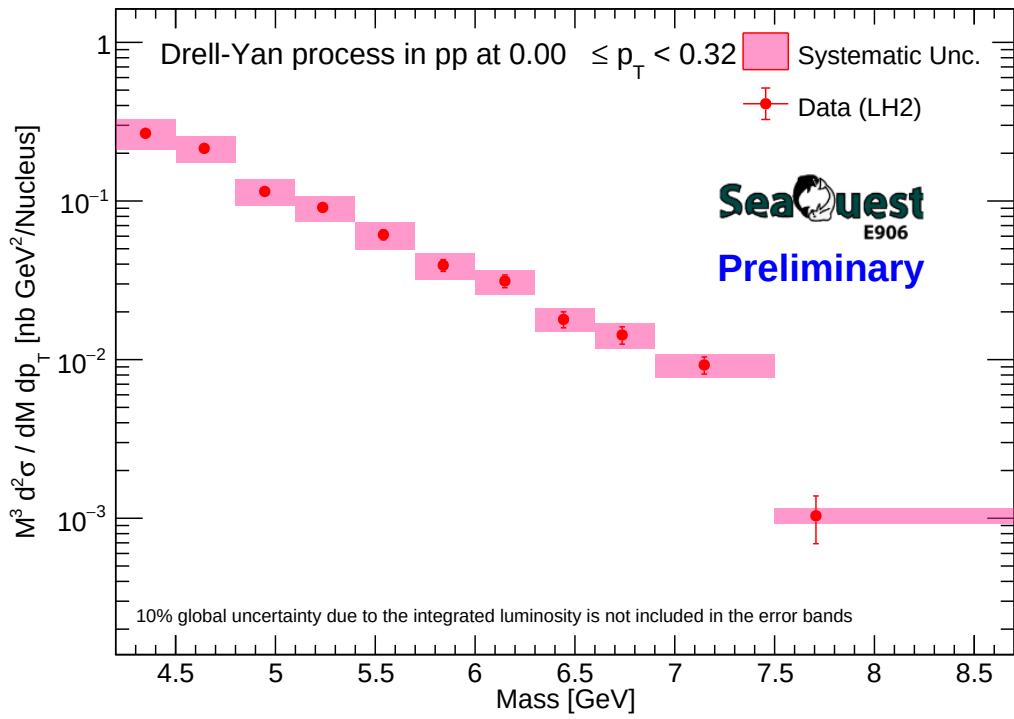


Figure 33: CrossSection_LH2_pT_0.00_0.32_Centroid_with_logo

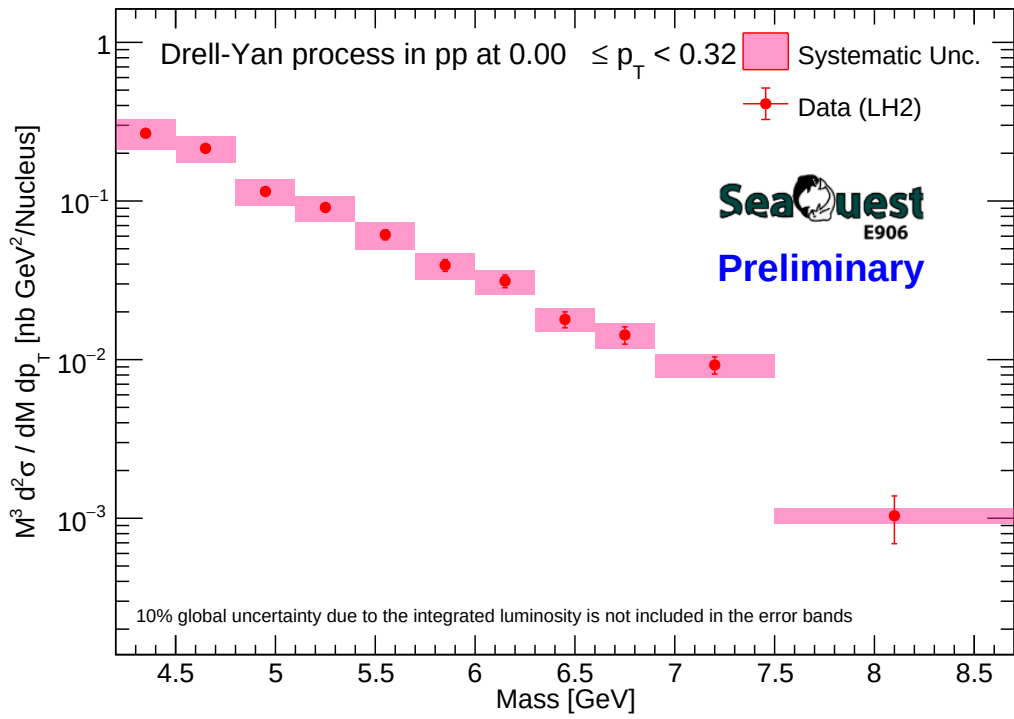


Figure 34: CrossSection_LH2_pT_0.00_0.32_GeoCenter_with_logo

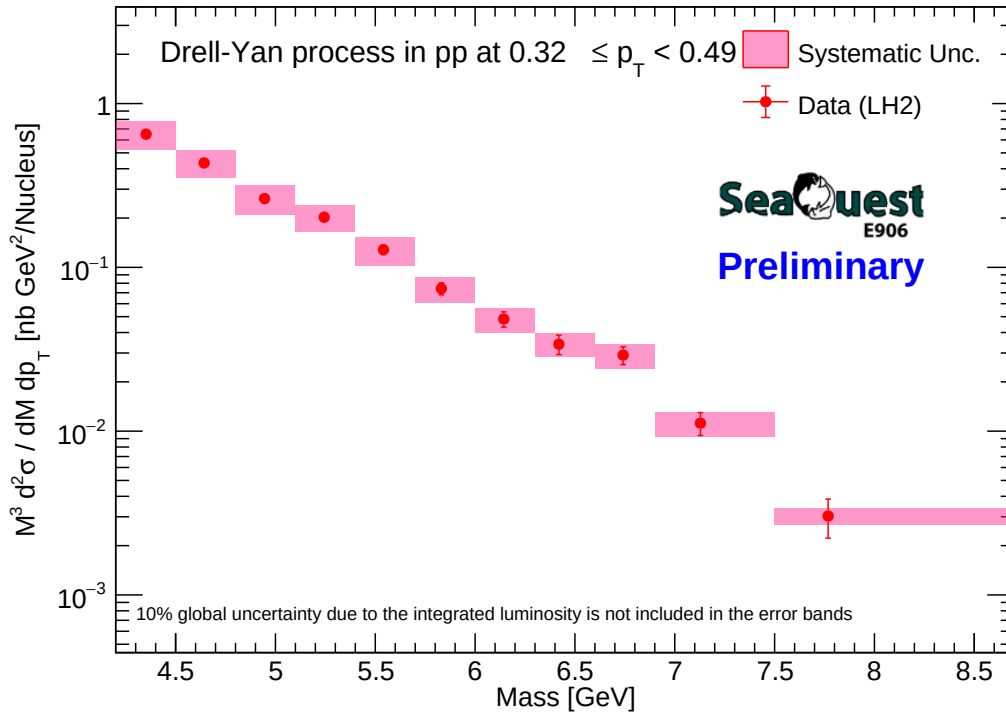


Figure 35: CrossSection_LH2_pT_0.32_0.49_Centroid_with_logo

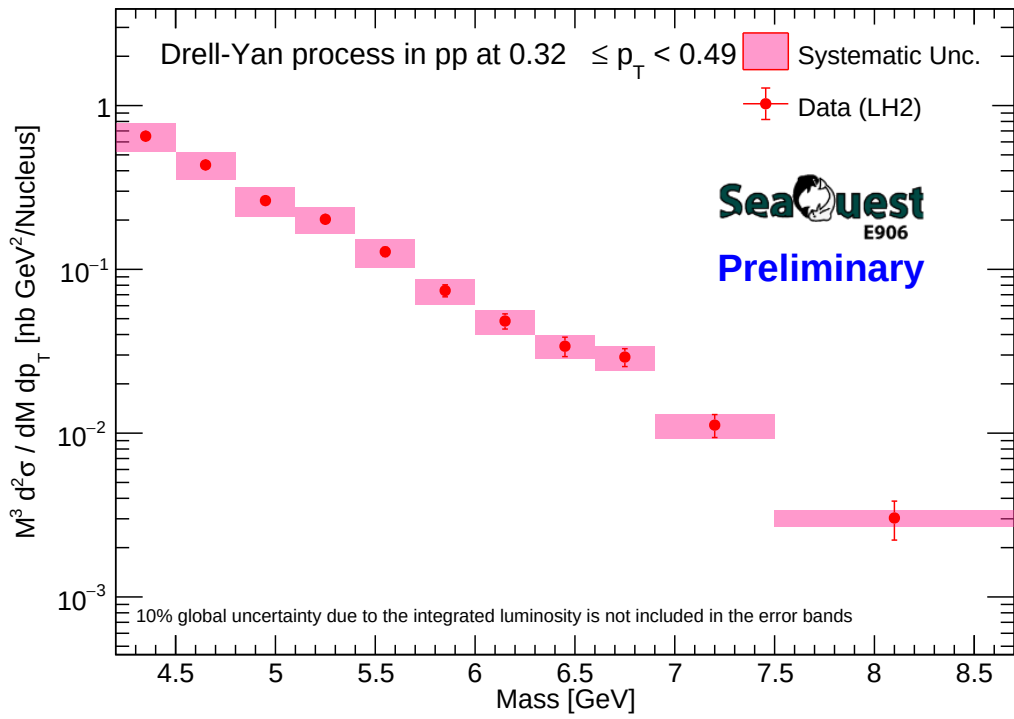


Figure 36: CrossSection_LH2_pT_0.32_0.49_GeoCenter_with_logo

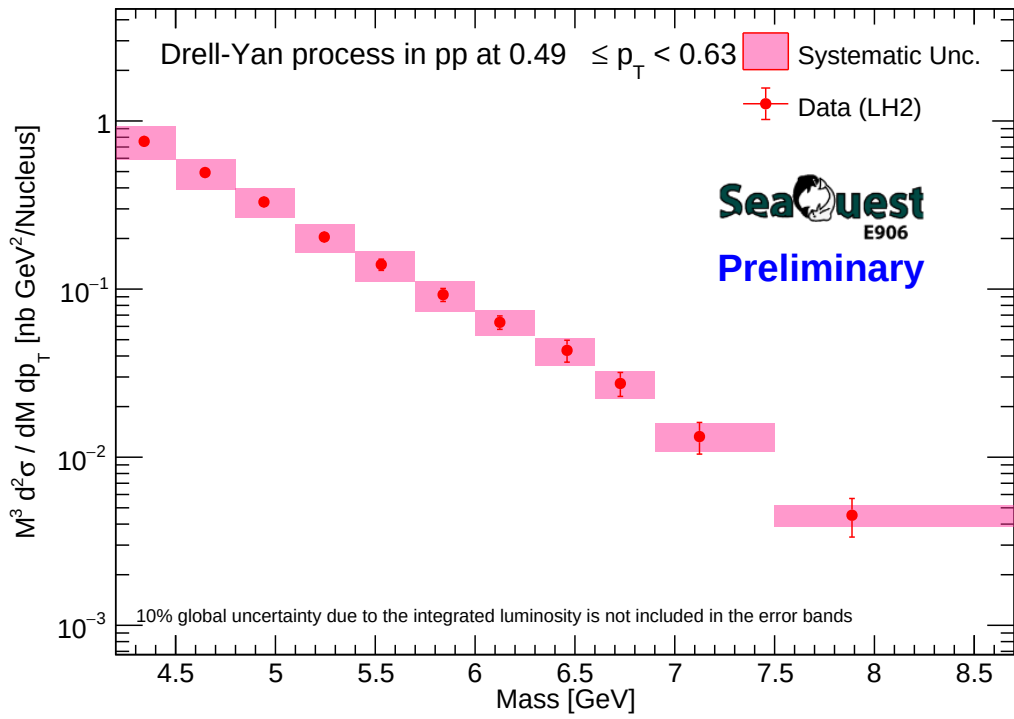


Figure 37: CrossSection_LH2_pT_0.49_0.63_Centroid_with_logo

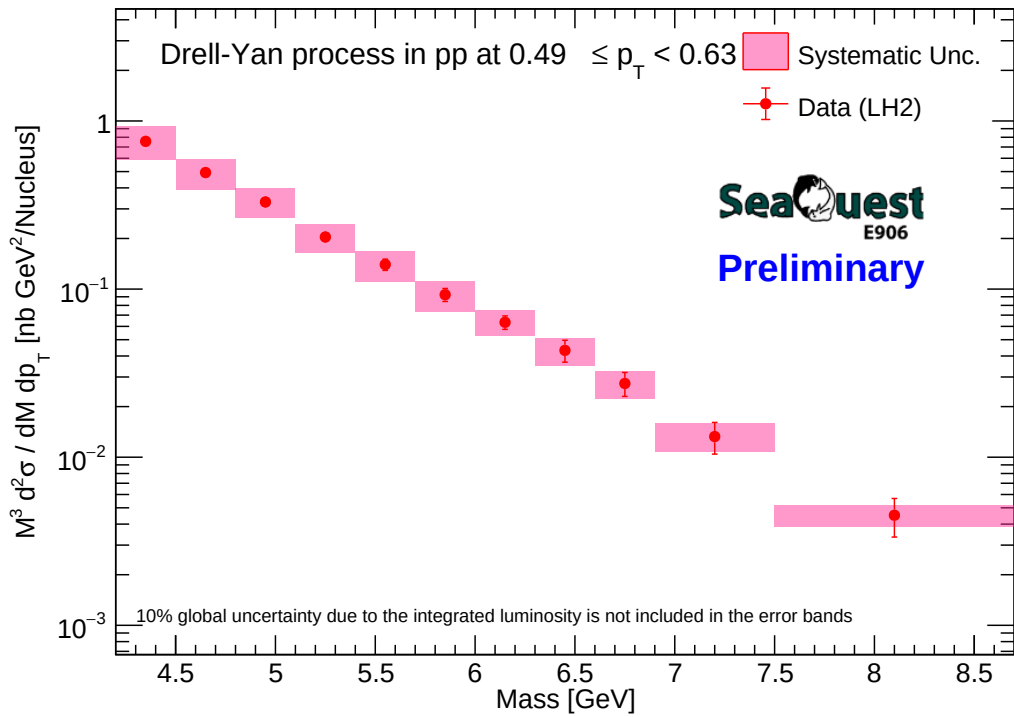


Figure 38: CrossSection_LH2_pT_0.49_0.63_GeoCenter_with_logo

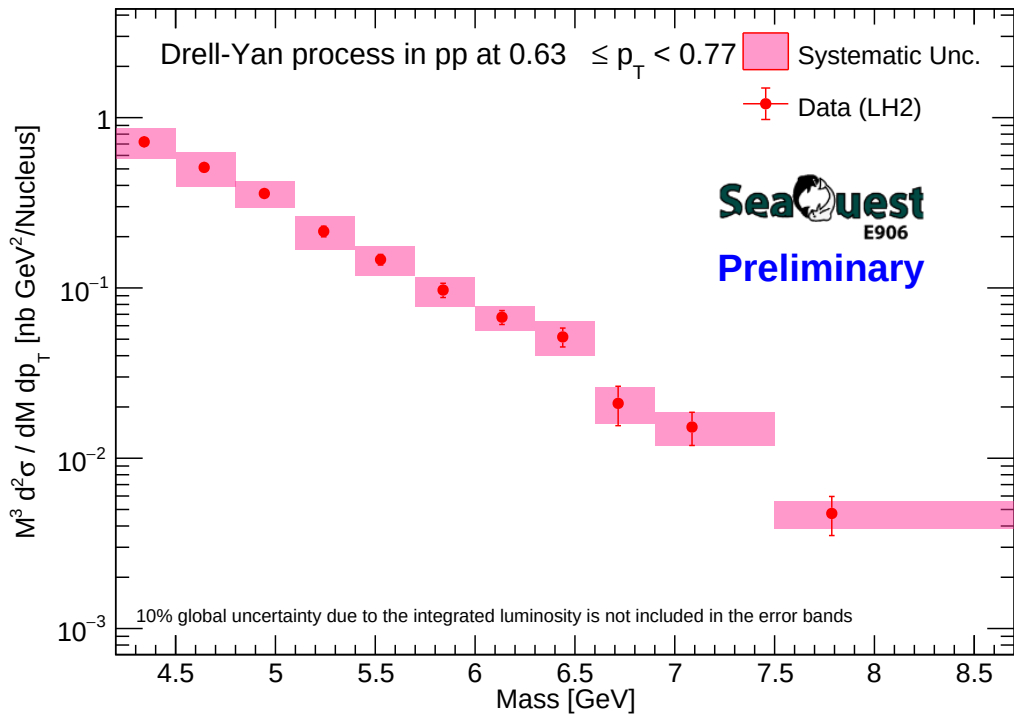


Figure 39: CrossSection_LH2_pT_0.63_0.77_Centroid_with_logo

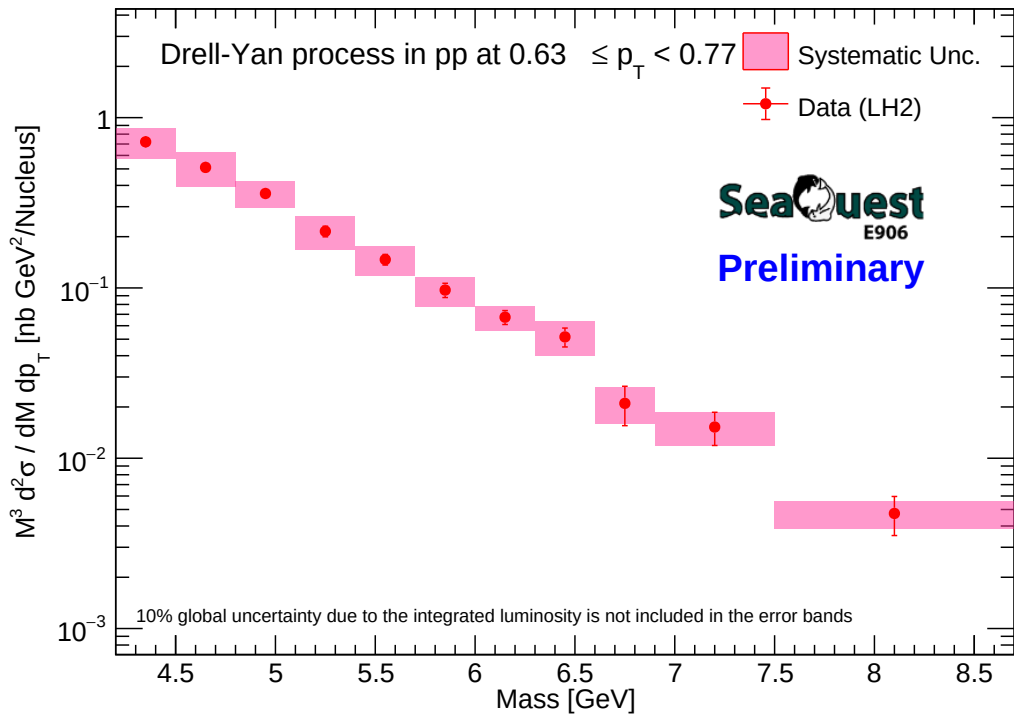


Figure 40: CrossSection_LH2_pT_0.63_0.77_GeoCenter_with_logo

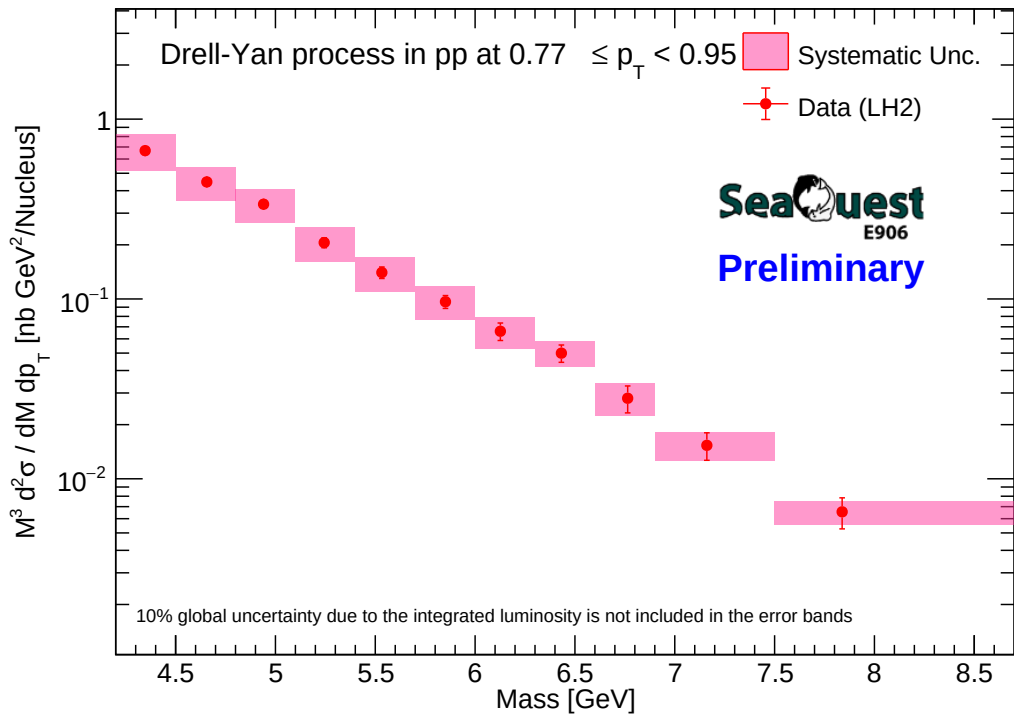


Figure 41: CrossSection_LH2_pT_0.77_0.95_Centroid_with_logo

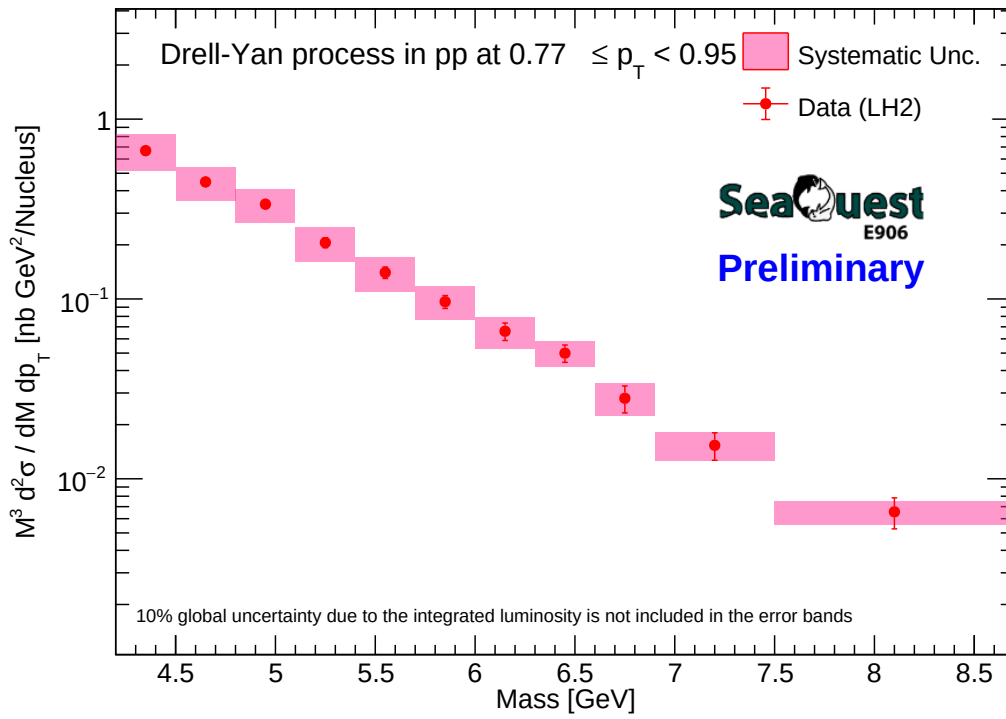


Figure 42: CrossSection_LH2_pT_0.77_0.95_GeoCenter_with_logo

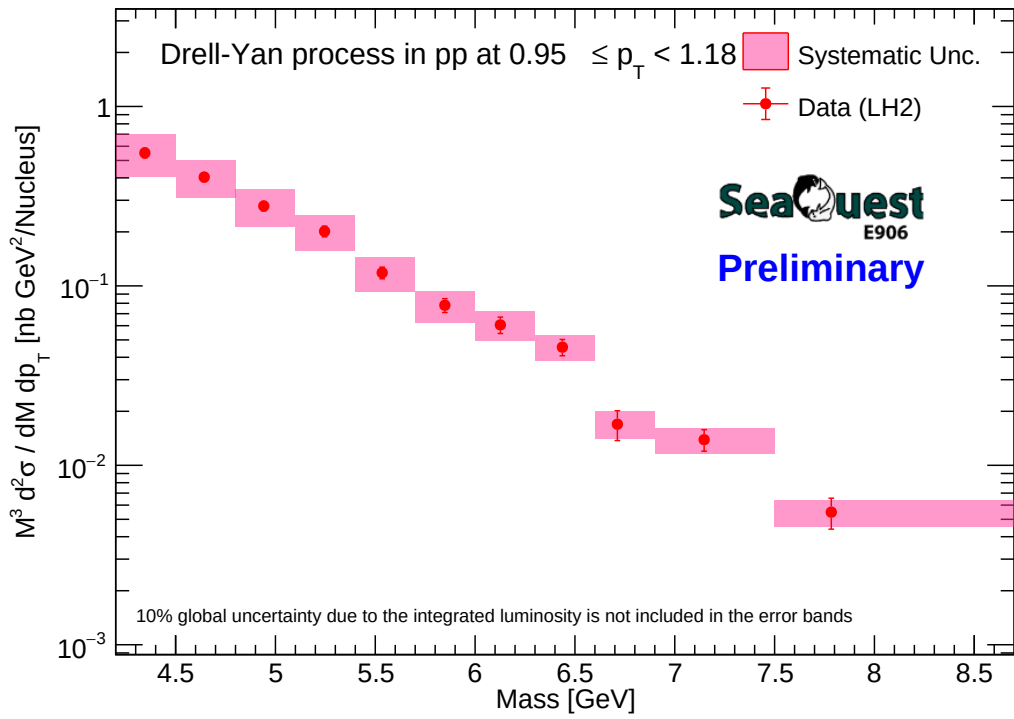


Figure 43: CrossSection_LH2_pT_0.95_1.18_Centroid_with_logo

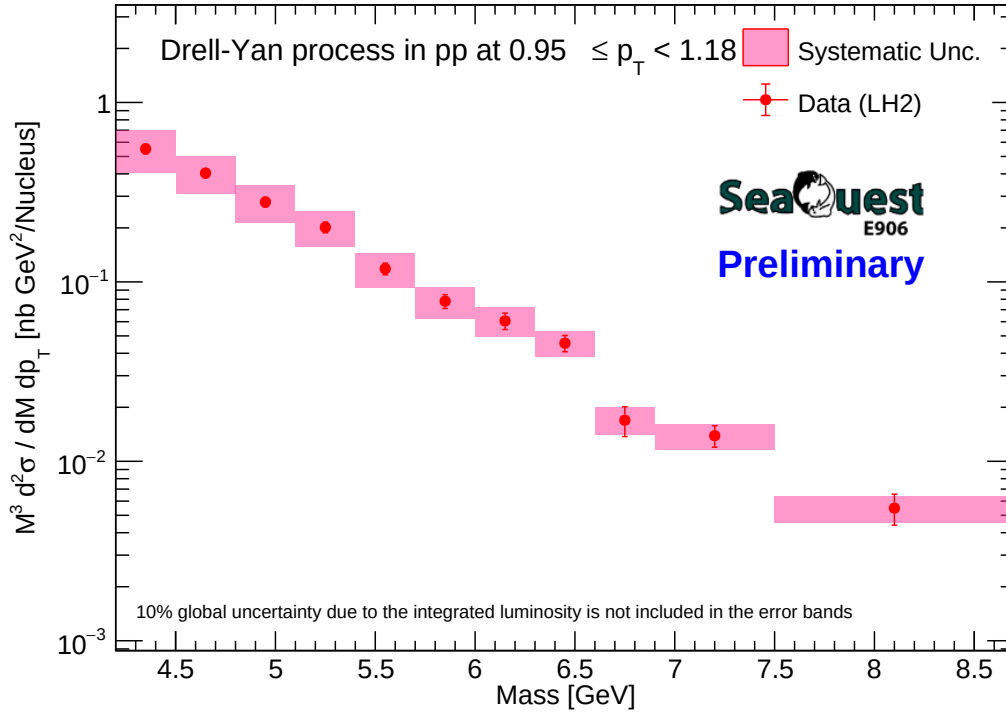


Figure 44: CrossSection_LH2_pT_0.95_1.18_GeoCenter_with_logo

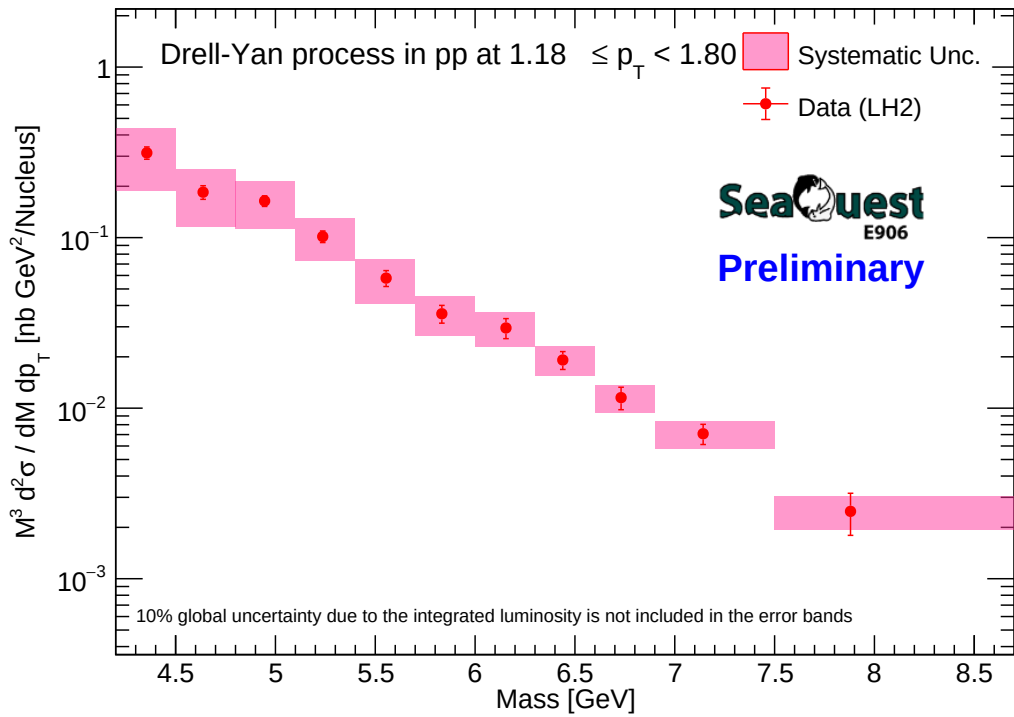


Figure 45: CrossSection_LH2_pT_1.18_1.80_Centroid_with_logo

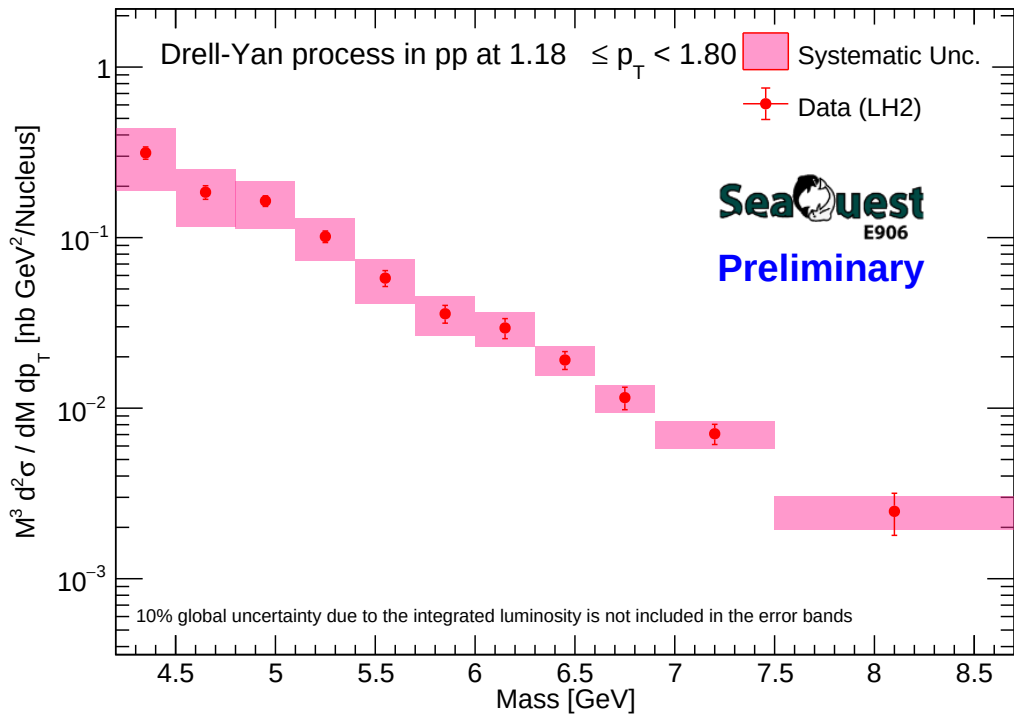


Figure 46: CrossSection_LH2_pT_1.18_1.80_GeoCenter_with_logo

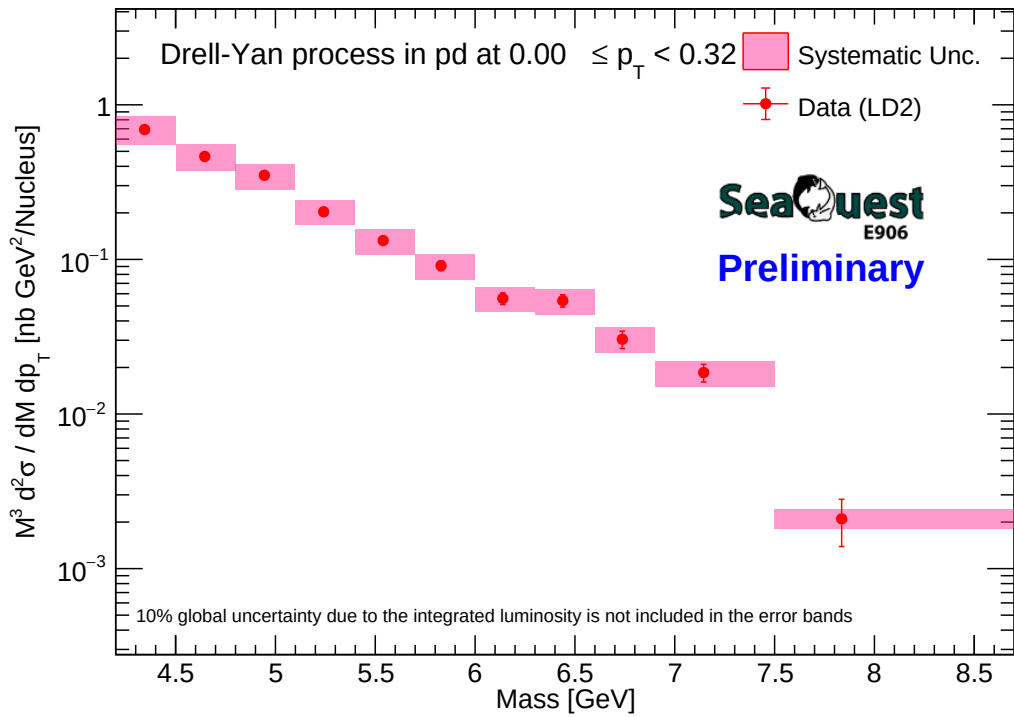


Figure 47: CrossSection_LD2_pT_0.00_0.32_Centroid_with_logo

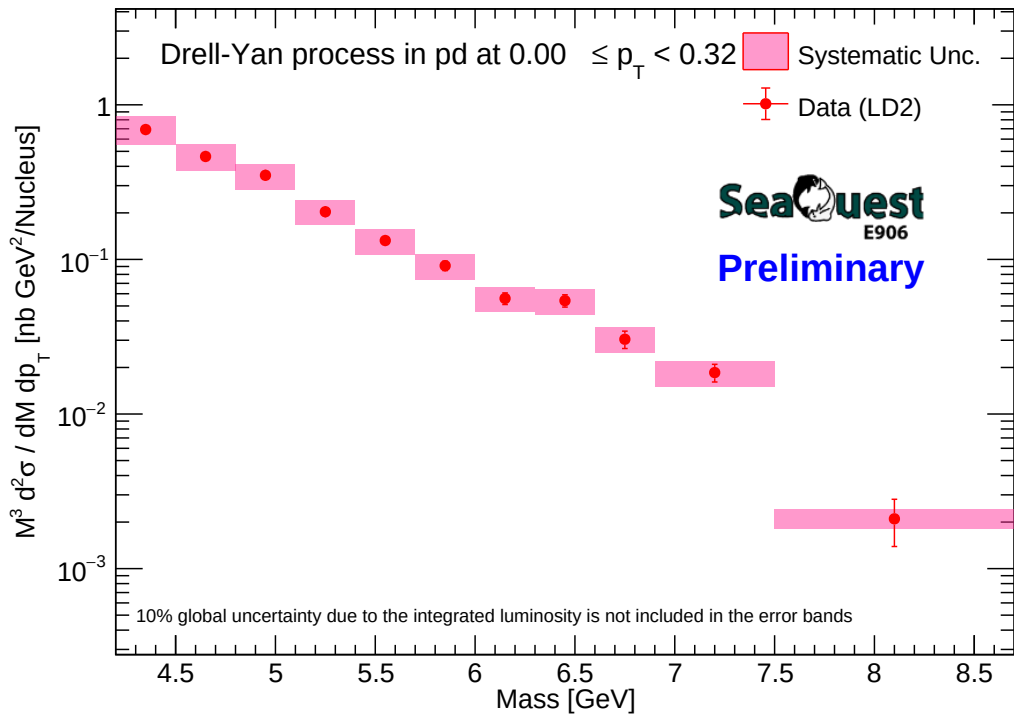


Figure 48: CrossSection_LD2_pT_0.00_0.32_GeoCenter_with_logo

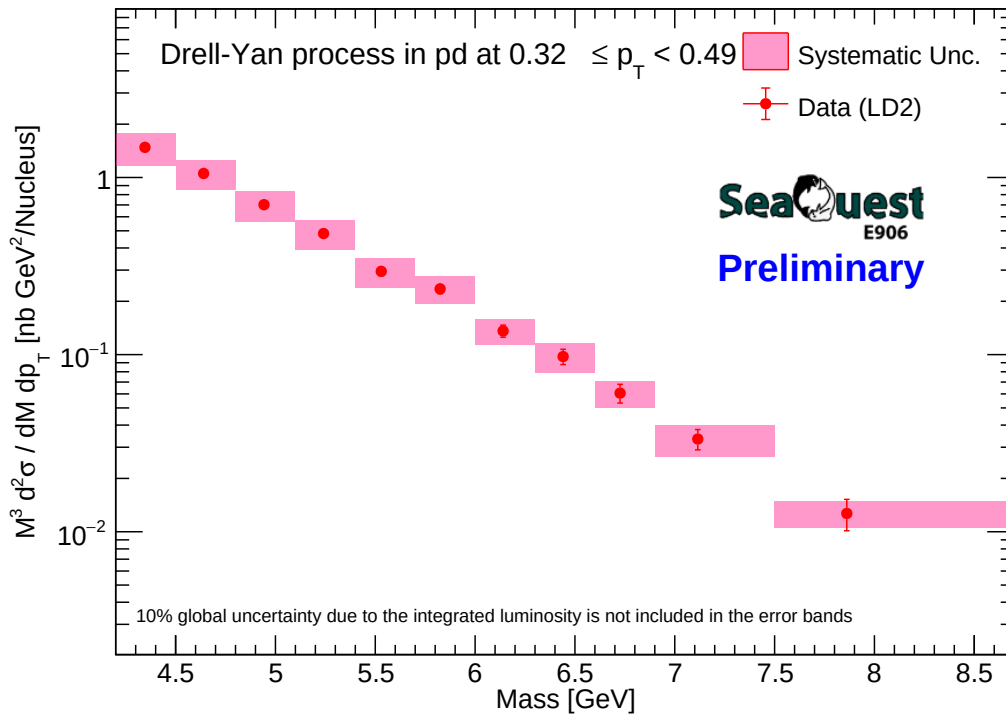


Figure 49: CrossSection_LD2_pT_0.32_0.49_Centroid_with_logo

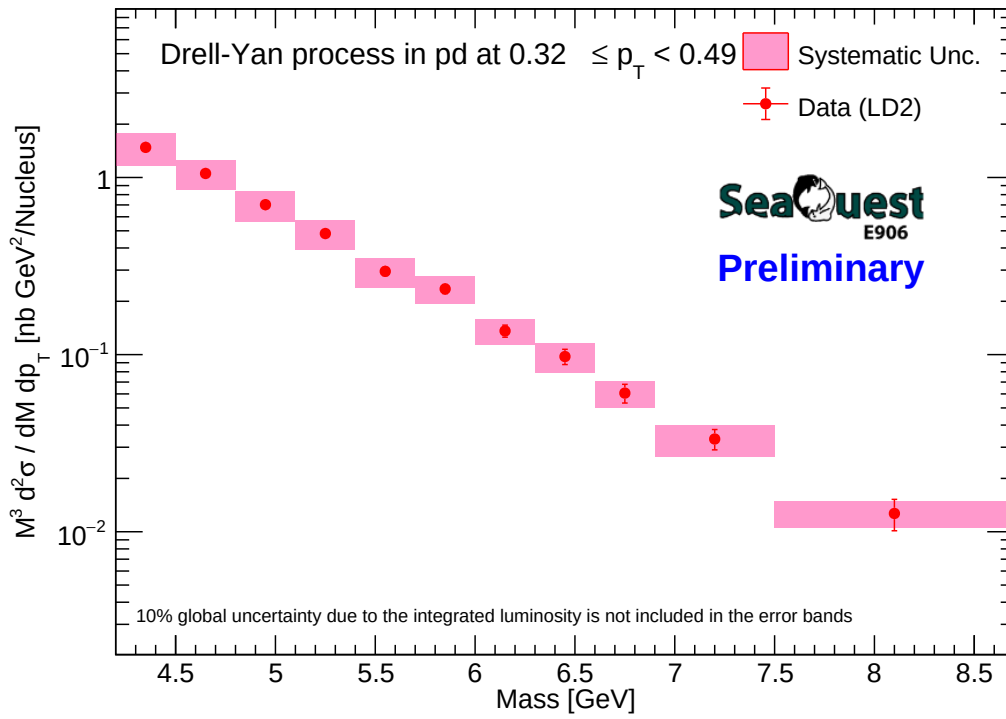


Figure 50: CrossSection_LD2_pT_0.32_0.49_GeoCenter_with_logo

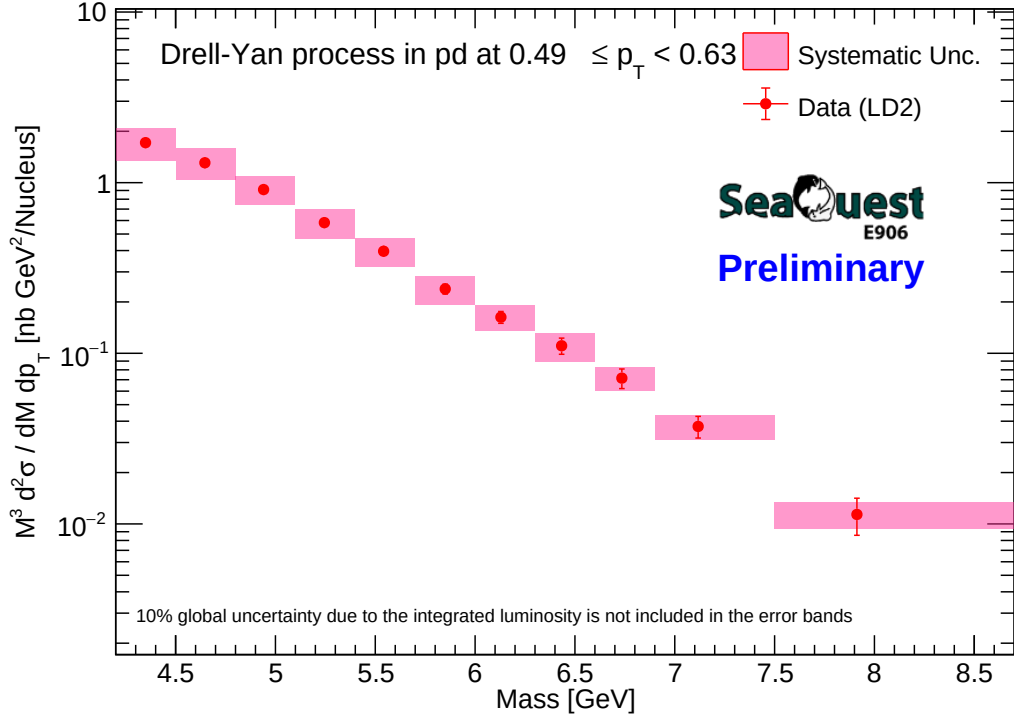


Figure 51: CrossSection_LD2_pT_0.49_0.63_Centroid_with_logo

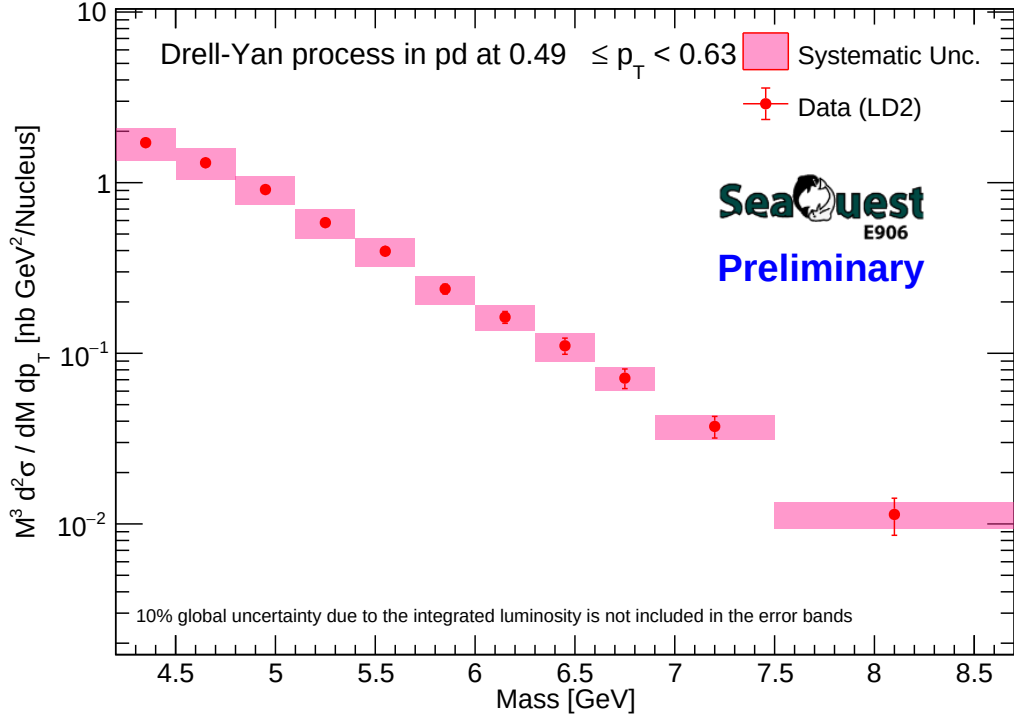


Figure 52: CrossSection_LD2_pT_0.49_0.63_GeoCenter_with_logo

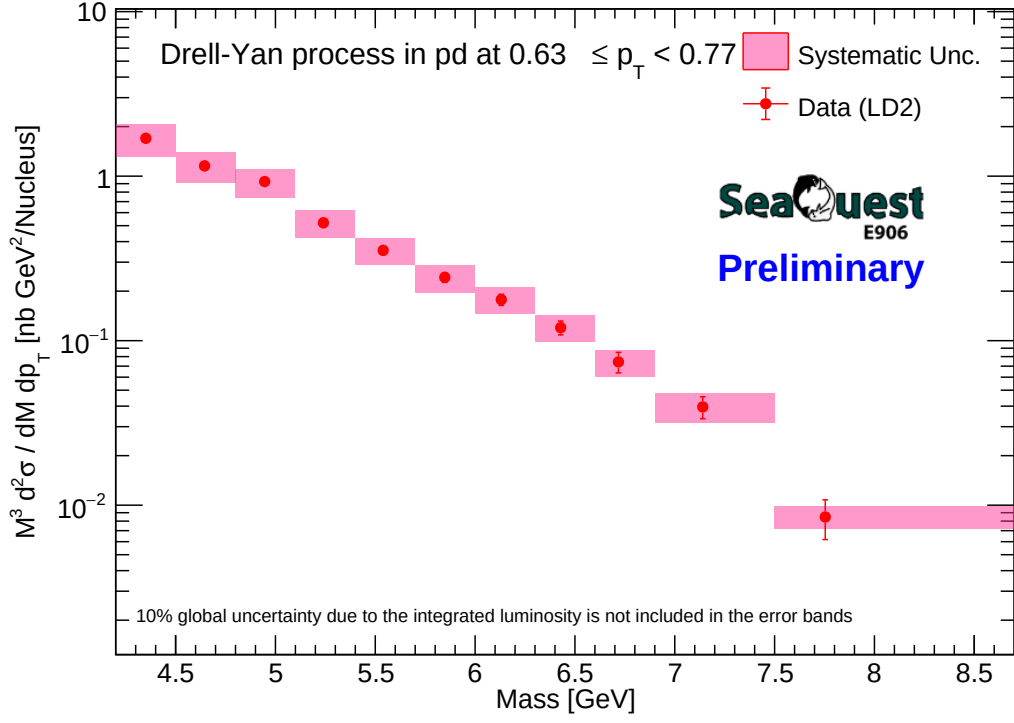


Figure 53: CrossSection_LD2_pT_0.63_0.77_Centroid_with_logo

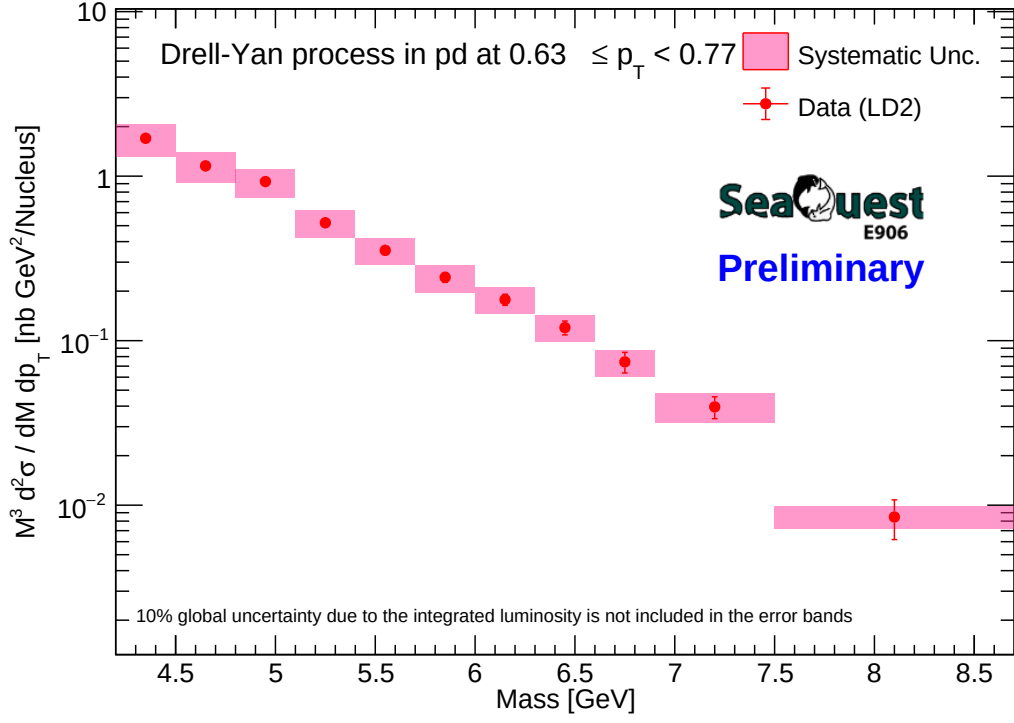


Figure 54: CrossSection_LD2_pT_0.63_0.77_GeoCenter_with_logo

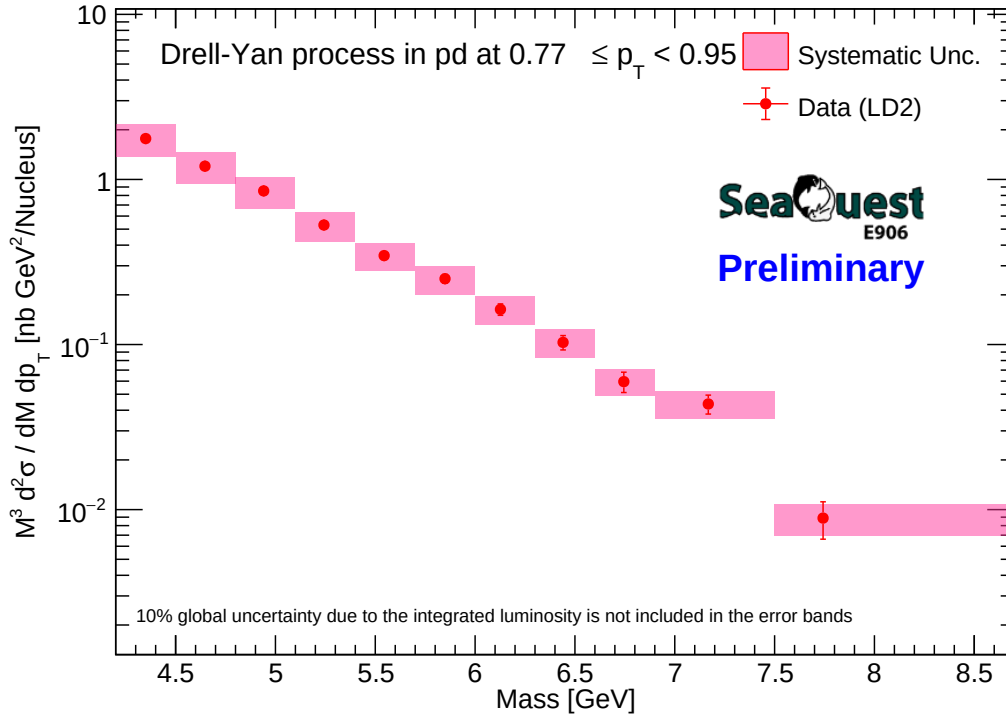


Figure 55: CrossSection_LD2_pT_0.77_0.95_Centroid_with_logo

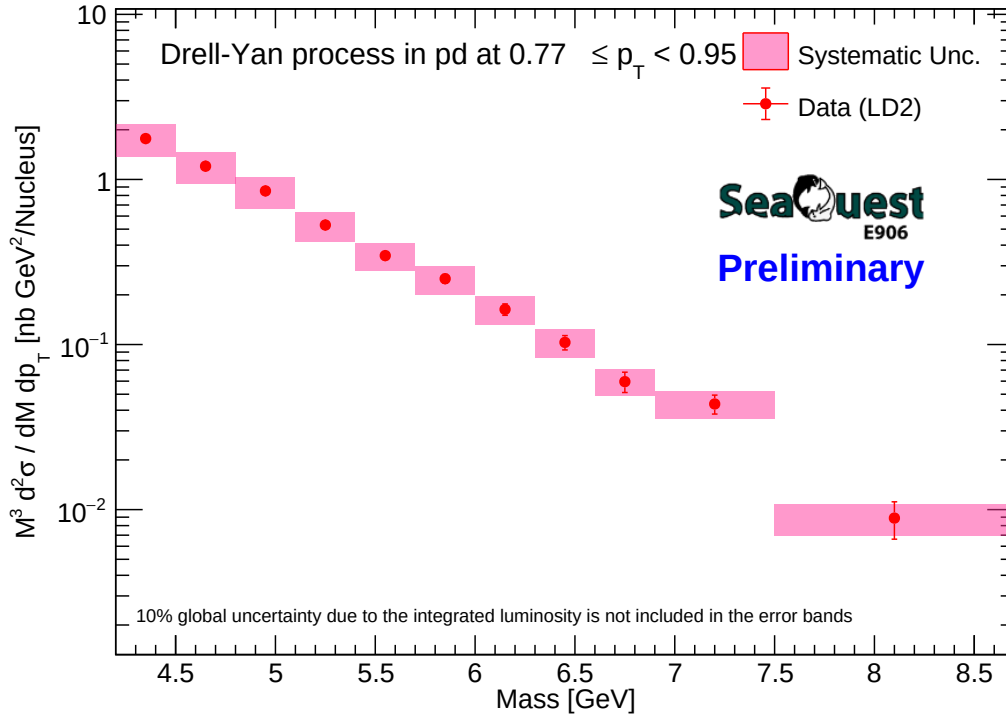


Figure 56: CrossSection_LD2_pT_0.77_0.95_GeoCenter_with_logo

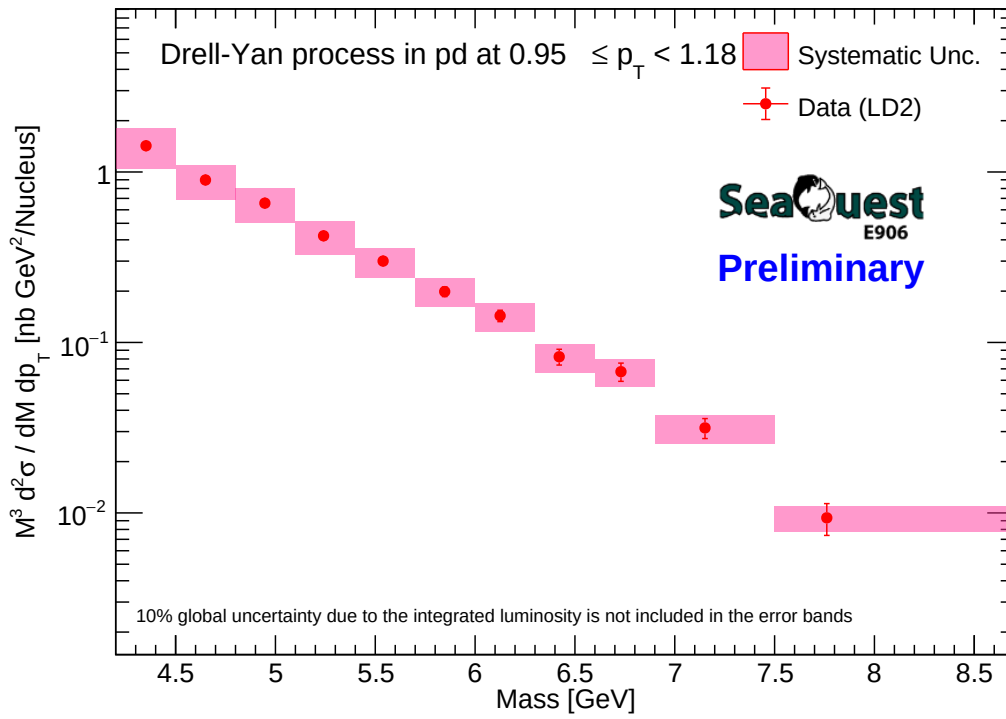


Figure 57: CrossSection_LD2_pT_0.95_1.18_Centroid_with_logo

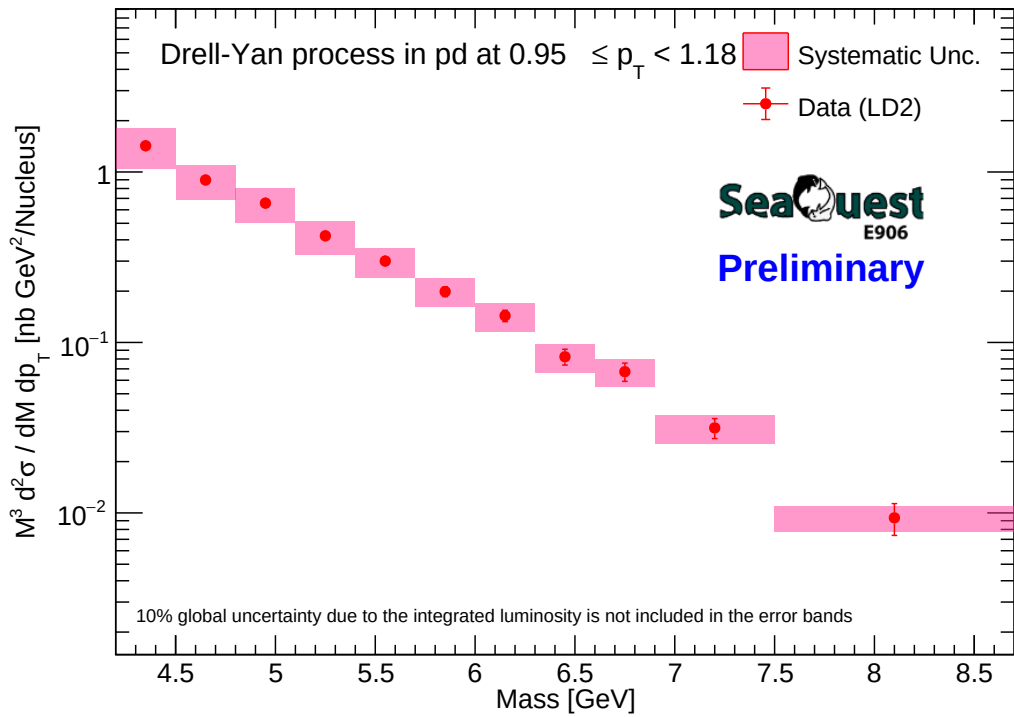


Figure 58: CrossSection_LD2_pT_0.95_1.18_GeoCenter_with_logo

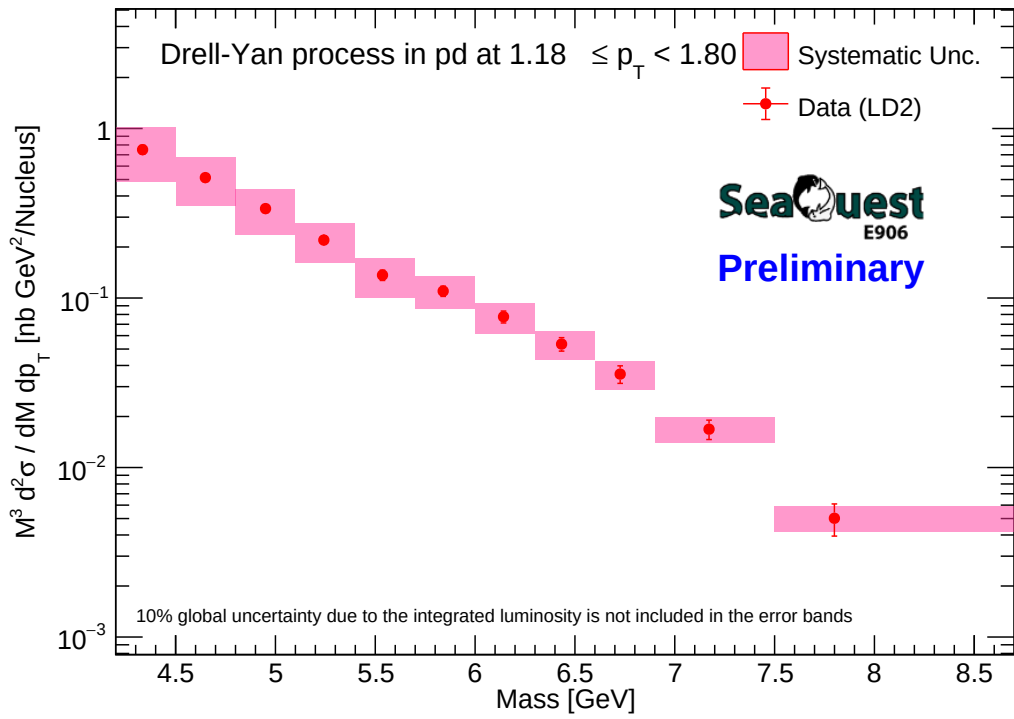


Figure 59: CrossSection_LD2_pT_1.18_1.80_Centroid_with_logo

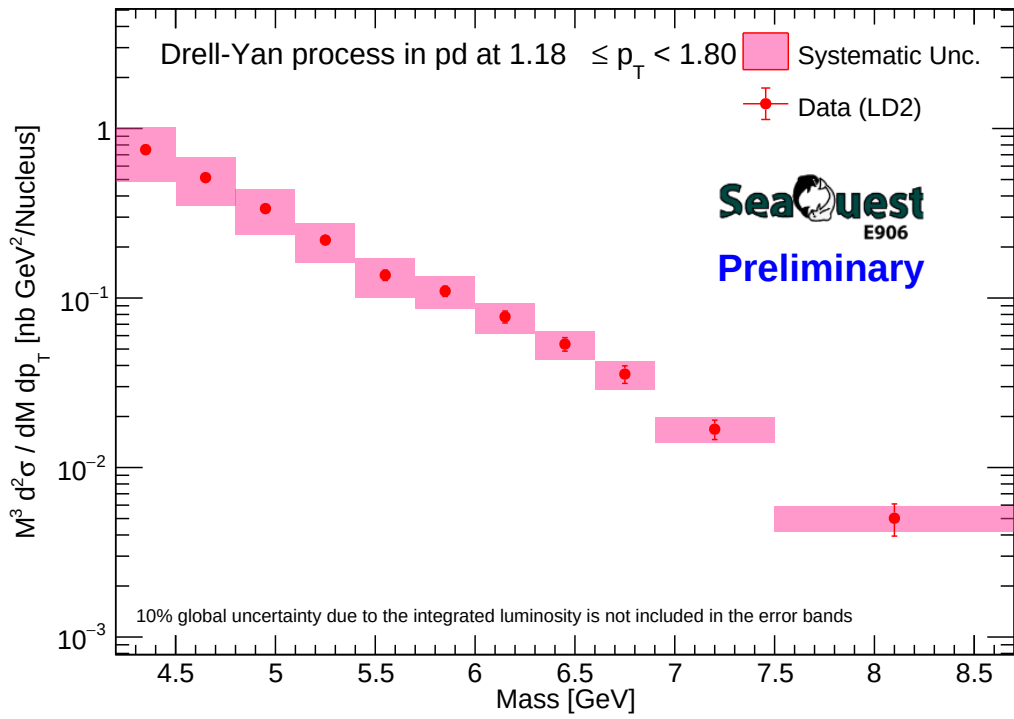


Figure 60: CrossSection_LD2_pT_1.18_1.80_GeoCenter_with_logo

DY Absolute Cross-Section Vs Mass for p_T bins (LD2)

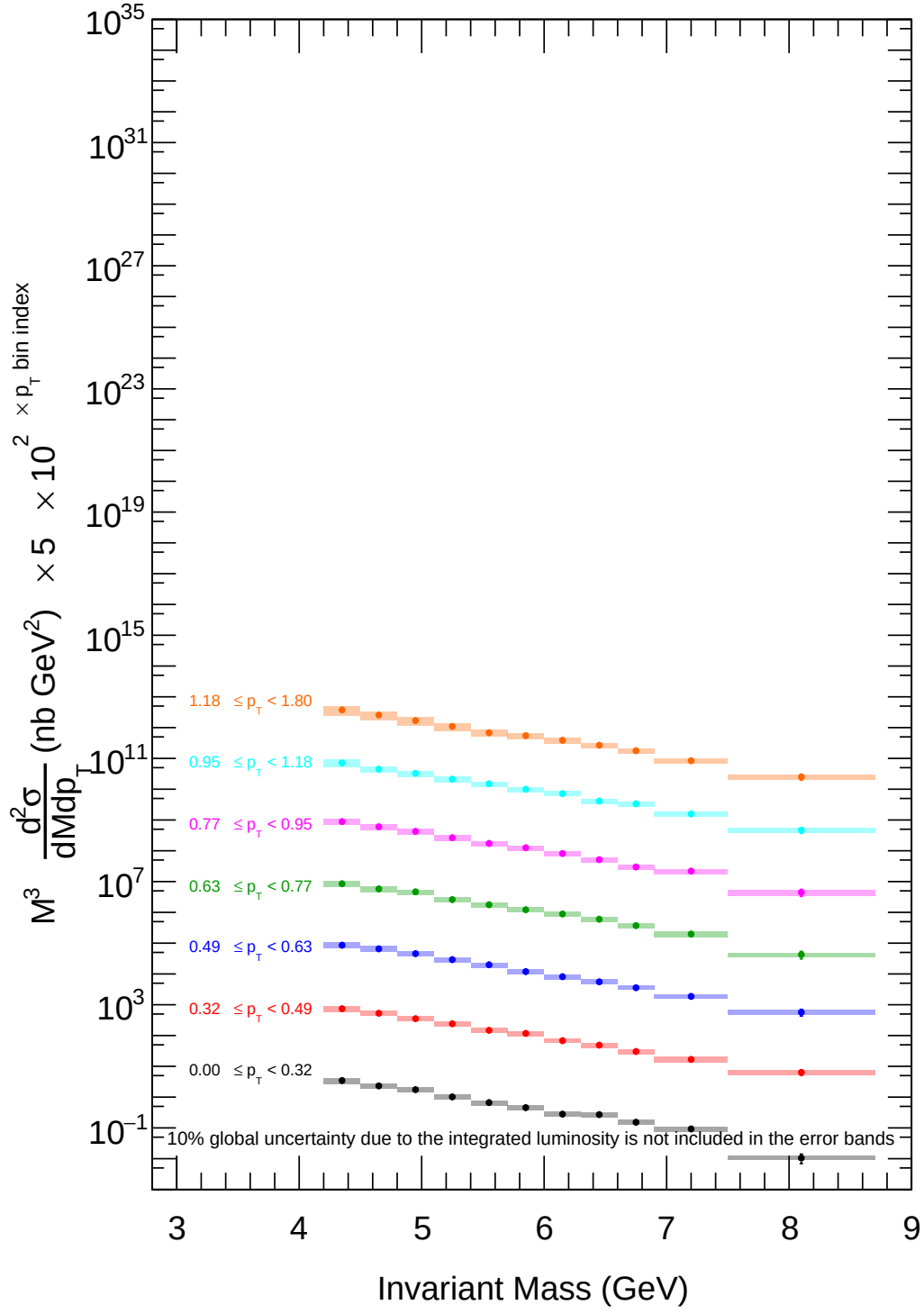


Figure 61: cross_section_overlay_LD2_GeoCenter